

Commutative Algebra

Lecture notes

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Tablet reading edition

The material draws substantially from David Eisenbud's *Commutative Algebra with a View Toward Algebraic Geometry* and Winfried Bruns and Jürgen Herzog's *Cohen–Macaulay Rings*.

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1 Modules

Definition. An A -module is an abelian group M with a scalar multiplication $A \times M \rightarrow M$ satisfying

$$(ab)m = a(bm), \quad 1m = m, \quad (a+b)m = am + bm, \quad a(m+n) = am + an.$$

A morphism $\varphi : M \rightarrow N$ is a group homomorphism such that $\varphi(am) = a\varphi(m)$. The category $\mathbf{Mod}_A$ is abelian.

Example. $\mathbf{Mod}_{\mathbb{Z}} = \mathbf{Ab}$; for a field k , $\mathbf{Mod}_k = \mathbf{Vect}_k$. The submodules of A are its ideals. Products and direct sums of A -modules are again A -modules. Write

$$M^I = \prod_{i \in I} M, \quad M^{(I)} = \bigoplus_{i \in I} M.$$

In particular, $A^{(I)}$ is the free A -module on I .

For $S \subseteq M$, the submodule it generates is

$$\langle S \rangle = \left\{ \sum_{s \in S} a_s s : a_s \in A, \text{ finitely many } a_s \neq 0 \right\}.$$

M is *finitely generated* if $M = \langle S \rangle$ for some finite S . It is *finitely presented* if there is an exact sequence

$$A^m \longrightarrow A^n \longrightarrow M \longrightarrow 0 \quad (m, n \in \mathbb{N}).$$

The group $\mathrm{Hom}_A(M, N)$ is an A -module under pointwise addition and scalar multiplication. Both $\mathrm{Hom}_A(M, -)$ and $\mathrm{Hom}_A(-, M)$ are left exact.

Exercise. If $M' \rightarrow M \rightarrow M''$ induces an exact sequence $\mathrm{Hom}_A(P, M') \rightarrow \mathrm{Hom}_A(P, M) \rightarrow \mathrm{Hom}_A(P, M'')$ for every P , then the original sequence is exact.

2 Algebras

Definition. An A -algebra is a ring B with a homomorphism $\varphi_B : A \rightarrow B$. An A -algebra morphism $\psi : B \rightarrow C$ makes the following triangle commute:

$$\begin{array}{ccc} & & B \\ & \nearrow \varphi_B & \downarrow \psi \\ A & & \\ & \searrow \varphi_C & \\ & & C \end{array}$$

These form the category $\mathbf{Alg}_A$. The forgetful functor to $\mathbf{Mod}_A$ uses $a \cdot b = \varphi_B(a)b$.

Example. For a commutative monoid P , the monoid algebra is $A[P] = A^{(P)}$, with multiplication $\delta_p \delta_q = \delta_{pq}$ and identity δ_1 . If P is the free commutative monoid on $x_1, \dots, x_n$, this is $A[x_1, \dots, x_n]$. Quotients A/I and products of A -algebras are A -algebras.

For $S \subseteq B$, write $A[S]$ for the generated A -subalgebra. The algebra B is *finitely generated* if $B = A[S]$ for finite S , and *finitely presented* if

$$B \simeq A[x_1, \dots, x_n]/I$$

for a finitely generated ideal I .

Regular elements and annihilators

An element $a \in A$ is M -regular, or a nonzerodivisor on M , if multiplication $M \xrightarrow{a} M$ is injective. Otherwise it is a zerodivisor on M . For $S \subseteq M$,

$$\mathrm{Ann}_A(S) = \{a \in A : as = 0 \text{ for all } s \in S\}.$$

3 Tensor products

Definition. The tensor product $M \otimes_A N$, with its bilinear map $(m, n) \mapsto m \otimes n$, represents bilinear maps: every A -bilinear map $b : M \times N \rightarrow P$ factors uniquely through an A -linear map $M \otimes_A N \rightarrow P$.

Construction. Take the free module $A^{(M \times N)}$ modulo the submodule generated by

$$\begin{aligned} (m + m', n) - (m, n) - (m', n), \\ (m, n + n') - (m, n) - (m, n'), \\ (am, n) - a(m, n), \quad (m, an) - a(m, n). \end{aligned}$$

The universal property proves uniqueness up to unique compatible isomorphism. More generally, tensor products of families of modules exist. $\square$

Proposition (Tensor–Hom adjunction). *There is a natural isomorphism*

$$\mathrm{Hom}_A(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P)).$$

It sends φ to $[m \mapsto (n \mapsto \varphi(m \otimes n))]$; the inverse sends ψ to $[m \otimes n \mapsto \psi(m)(n)]$.

Thus $- \otimes_A N$ is a left adjoint and preserves colimits; in particular it is right exact. **Exercises.** Establish the natural isomorphisms

$$\begin{aligned} (M \otimes_A N) \otimes_A P &\simeq M \otimes_A (N \otimes_A P), \\ A \otimes_A M &\simeq M \otimes_A A \simeq M, \quad M \otimes_A N \simeq N \otimes_A M, \\ (\mathrm{colim}_i M_i) \otimes_A N &\simeq \mathrm{colim}_i (M_i \otimes_A N). \end{aligned}$$

4 Change of rings

Definition. Fix $\varphi : A \rightarrow B$. Restriction of scalars sends $N \in \mathbf{Mod}_B$ to N_A , where $a \cdot n = \varphi(a)n$. Extension of scalars sends $M \in \mathbf{Mod}_A$ to $M_B = M \otimes_A B$, where $b(m \otimes b') = m \otimes bb'$. For a map u , write $u_B = u \otimes_A B$.

Proposition. *There is a natural isomorphism*

$$\mathrm{Hom}_B(M \otimes_A B, N) \simeq \mathrm{Hom}_A(M, N_A).$$

It sends f to $[m \mapsto f(m \otimes 1)]$, with inverse $g \mapsto [m \otimes b \mapsto bg(m)]$. Thus extension and restriction of scalars form an adjoint pair.

We often omit the subscript in N_A . If N is a B -module, then $M \otimes_A N$ is a B -module by $b(m \otimes n) = m \otimes bn$. For $A \rightarrow B \rightarrow C$, $M \in \mathbf{Mod}_A$, $N \in \mathbf{Mod}_B$, and $P \in \mathbf{Mod}_C$, there is a natural C -linear isomorphism

$$(M \otimes_A N) \otimes_B P \simeq M \otimes_A (N \otimes_B P),$$

sending $(m \otimes n) \otimes p$ to $m \otimes (n \otimes p)$.

Proposition. *For A -algebras B, C , the algebra $B \otimes_A C$ is their coproduct in $\mathbf{Alg}_A$:*

$$\mathrm{Hom}_{\mathbf{Alg}_A}(B \otimes_A C, D) \simeq \mathrm{Hom}_{\mathbf{Alg}_A}(B, D) \times \mathrm{Hom}_{\mathbf{Alg}_A}(C, D).$$

The inverse sends (f, g) to $[b \otimes c \mapsto f(b)g(c)]$.

5 Flat modules

Definition. An A -module M is *flat* if $- \otimes_A M$ is exact. An A -algebra, or a homomorphism $A \rightarrow B$, is flat if its underlying A -module is flat. **Examples and exercises.**

1. Free modules and their direct summands are flat. Direct sums and filtered direct limits of flat modules are flat.
2. A flat module is A -torsion-free: every A -regular element is M -regular.
3. If M is flat over A , then $M \otimes_A B$ is flat over B .
4. If $A \rightarrow B$ is flat and N is flat over B , then N_A is flat. In particular, composites of flat homomorphisms are flat.

Proposition. *For $A \rightarrow B$ there is a natural map*

$$\Phi_{M,N} : \operatorname{Hom}_A(M, N) \otimes_A B \longrightarrow \operatorname{Hom}_B(M_B, N_B), \quad f \otimes b \longmapsto bf_B.$$

It is an isomorphism if B is flat over A and M is finitely presented.

Proof. It is an isomorphism for $M = A^n$, without flatness. Choose a presentation $A^m \rightarrow A^n \rightarrow M \rightarrow 0$. Abbreviate $H(X) = \operatorname{Hom}_A(X, N) \otimes_A B$ and $K(X) = \operatorname{Hom}_B(X_B, N_B)$. Naturality gives

$$\begin{array}{ccccccc} 0 & \longrightarrow & H(M) & \longrightarrow & H(A^n) & \longrightarrow & H(A^m) \\ & & \downarrow \Phi_M & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & K(M) & \longrightarrow & K(A^n) & \longrightarrow & K(A^m). \end{array}$$

The top row is exact by flatness, and the bottom by left exactness of Hom . Hence the first vertical map is an isomorphism. $\square$

6 The prime spectrum

Definition. Write $\text{Ideal}(A)$ for the ideals of A , $\text{Spec } A$ for its prime ideals, and $\text{Max } A$ for its maximal ideals. For $\varphi : A \rightarrow B$, extension and contraction are

$$IB = (\varphi(I))B, \quad A \cap J = \varphi^{-1}(J).$$

For $E \subseteq A$ and $f \in A$, put

$$V(E) = \{\mathfrak{p} \in \text{Spec } A : E \subseteq \mathfrak{p}\},$$

$$D(f) = (\text{Spec } A)_f = \{\mathfrak{p} : f \notin \mathfrak{p}\}, \quad (\text{Spec } A)_S = \bigcap_{f \in S} D(f).$$

Exercises. If $(E) = I$, then $V(E) = V(I)$. Moreover,

$$V(I) \cup V(J) = V(I \cap J), \quad \bigcap_{\alpha} V(I_{\alpha}) = V\left(\sum_{\alpha} I_{\alpha}\right).$$

These sets are precisely the closed sets of the *Zariski topology*.

Proposition. A homomorphism $\varphi : A \rightarrow B$ induces a continuous map $\varphi^* : \text{Spec } B \rightarrow \text{Spec } A$, $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$.

Proof. $(\varphi^*)^{-1}(V_A(E)) = V_B(\varphi(E))$. □

Example. $\text{Spec } \mathbb{Z} = \{(0)\} \cup \{(p) : p \text{ a positive prime}\}$. For a field k , $\text{Spec } k[x]$ consists of (0) and ideals generated by irreducible polynomials.

Example. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[\sqrt{5}]$ commutes with the automorphism $\sigma(\sqrt{5}) = -\sqrt{5}$ upstairs. Its induced action on the spectrum fixes the prime $(2, 1 + \sqrt{5}) = (2, 1 - \sqrt{5})$ lying over (2) . The ring maps commute:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\quad} & \mathbb{Z}[\sqrt{5}] \\ & \searrow & \downarrow \sigma \\ & & \mathbb{Z}[\sqrt{5}]. \end{array}$$

7 Localization

Definition. A subset $S \subseteq A$ is *multiplicative* if $1 \in S$ and $ss' \in S$ for all $s, s' \in S$. For an A -module M , define $M_S = (M \times S)/\sim$, where

$$(m, s) \sim (m', s') \iff \exists t \in S : t(s'm - sm') = 0.$$

Write m/s for the equivalence class and $\ell_S(m) = m/1$. The usual rules

$$\frac{m}{s} + \frac{m'}{s'} = \frac{s'm + sm'}{ss'}, \quad \frac{a}{s} \frac{m}{s'} = \frac{am}{ss'}$$

make A_S an A -algebra and M_S an A_S -module.

Proposition. A_S is universal among A -algebras $\varphi : A \rightarrow B$ for which $\varphi(S) \subseteq B^\times$.

Proof. The unique extension is $\varphi_S(a/s) = \varphi(a)\varphi(s)^{-1}$. Check that it is well defined and multiplicative. $\square$

If B is an A -algebra and $\bar{S} = \varphi(S)$, localization of its underlying A -module gives $B_S \simeq B_{\bar{S}}$.

1. For $S(f) = \{1, f, f^2, \dots\}$, write $M_f = M_{S(f)}$. Then $(M_f)_g \simeq M_{fg}$, and $\operatorname{colim}_{f \in S} M_f \simeq M_S$, with the natural transition maps under further localization.
2. If S is the set of nonzerodivisors of A , then $Q(A) = A_S$ is the *total ring of fractions*. For a domain this is the fraction field $\operatorname{Frac} A$.
3. For $\mathfrak{p} \in \operatorname{Spec} A$, write $M_{\mathfrak{p}} = M_{A \setminus \mathfrak{p}}$, the localization at $\mathfrak{p}$.

8 Localization of modules and local rings

Localization is a functor $\mathbf{Mod}_A \rightarrow \mathbf{Mod}_{A_S}$, sending φ to $\varphi_S(m/s) = \varphi(m)/s$.

Proposition. *Localization is left adjoint to restriction of scalars, is naturally isomorphic to $- \otimes_A A_S$, and is exact. Consequently A_S is flat over A .*

Proof. The adjunction is

$$\mathrm{Hom}_{A_S}(M_S, N) \simeq \mathrm{Hom}_A(M, N_A),$$

with inverse $g \mapsto [m/s \mapsto s^{-1}g(m)]$. Both localization and tensoring with A_S are left adjoint to restriction, so they are naturally isomorphic. They are therefore right exact. If $\varphi : M \hookrightarrow N$ and $\varphi_S(m/s) = 0$, some $t \in S$ satisfies $\varphi(tm) = 0$. Injectivity gives $tm = 0$, hence $m/s = 0$. Thus localization is also left exact. $\square$

For $N \subseteq M$, localization of $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$ gives

$$(M/N)_S \simeq M_S/N_S.$$

For $\mathfrak{p} \in \mathrm{Spec} A$,

$$\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \simeq (A/\mathfrak{p})_{\mathfrak{p}} \simeq \mathrm{Frac}(A/\mathfrak{p}).$$

The ideal $\mathfrak{p}A_{\mathfrak{p}}$ is the unique maximal ideal of $A_{\mathfrak{p}}$. A *local ring* $(A, \mathfrak{m}, k)$ has a unique maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. A local homomorphism $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ is a ring map with inverse image of $\mathfrak{n}$ equal to $\mathfrak{m}$.

9 Localization and submodules

For $\ell_S : M \rightarrow M_S$,

$$\ker \ell_S = \{m \in M : S \cap \text{Ann}_A(m) \neq \emptyset\}.$$

Thus ℓ_S is injective precisely when every $s \in S$ is M -regular.

Proposition. *For $P \subseteq M_S$ and $N \subseteq M$,*

$$(\ell_S^{-1}P)_S = P, \quad \ell_S^{-1}(N_S) = \{m \in M : sm \in N \text{ for some } s \in S\}.$$

Consequently localization is surjective on submodule posets. Its section is contraction, whose image consists of the N for which every $s \in S$ is (M/N) -regular.

Proof. The first identity follows by clearing a denominator. The second is the kernel formula for M/N , using $(M/N)_S \simeq M_S/N_S$. The final assertion says precisely that $M/N \rightarrow (M/N)_S$ is injective. $\square$

Corollary. *Contraction gives a homeomorphism*

$$\text{Spec } A_S \simeq \{\mathfrak{p} \in \text{Spec } A : \mathfrak{p} \cap S = \emptyset\} = (\text{Spec } A)_S.$$

Proof. The submodule correspondence gives the bijection on primes. For $J \subseteq A_S$ and $I = \ell_S^{-1}(J)$, we have $IA_S = J$ and

$$\begin{array}{ccc} A & \xrightarrow{\ell_S} & A_S \\ \downarrow & & \downarrow \\ A/I & \longrightarrow & (A/I)_S \simeq A_S/J. \end{array}$$

The image of $V(J)$ is $V(I) \cap (\text{Spec } A)_S$. Hence the continuous bijection is closed onto this subspace. $\square$

10 Radical ideals

Definition. Order ideals by inclusion and closed sets by reverse inclusion. For $Z \subseteq \operatorname{Spec} A$, set

$$I(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

The maps V and I preserve these orders, and $E \subseteq I(V(E))$, $Z \subseteq V(I(Z))$. For an ideal J ,

$$\sqrt{J} = \{a \in A : a^n \in J \text{ for some } n \geq 1\}.$$

J is *radical* if $J = \sqrt{J}$. The nilradical is $\sqrt{0}$; a ring is *reduced* if $\sqrt{0} = 0$.

Exercises. Radical is order preserving and idempotent;

$$\sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J} = \sqrt{IJ}.$$

Any intersection of radical ideals, in particular of prime ideals, is radical.

Proposition. $I(V(J)) = \sqrt{J}$ and $V(I(Z)) = \overline{Z}$. Thus radical ideals correspond bijectively to closed subsets of $\operatorname{Spec} A$.

Proof. For the first assertion pass to A/J . An element f lies in every prime precisely when $\operatorname{Spec} A_f$ is empty, equivalently $A_f = 0$, equivalently f is nilpotent. For the second, $Z \subseteq V(I(Z))$. If $Z \subseteq V(f)$, then $f \in I(Z)$, so $V(I(Z)) \subseteq V(f)$. Intersecting over such closed sets proves the assertion. $\square$

11 Local properties

Proposition. *For an A -module M ,*

$$M = 0 \iff M_{\mathfrak{p}} = 0 \ (\forall \mathfrak{p} \in \operatorname{Spec} A) \iff M_{\mathfrak{m}} = 0 \ (\forall \mathfrak{m} \in \operatorname{Max} A).$$

If $(f_1, \dots, f_n) = A$, then $M = 0$ if and only if every $M_{f_i} = 0$.

Proof. If $0 \neq m \in M$, choose a maximal ideal $\mathfrak{m} \supseteq \operatorname{Ann}(m)$. No element outside $\mathfrak{m}$ kills m , so $m/1 \neq 0$ in $M_{\mathfrak{m}}$. For the second assertion, if every $M_{f_i} = 0$, choose powers $f_i^{e_i}$ killing a given m . The ideal generated by these powers is the unit ideal, since its radical contains every f_i . Hence $m = 0$. $\square$

A property of rings is *local* if it is preserved by $A \mapsto A_f$ and can be checked on a finite principal cover: whenever $(f_1, \dots, f_n) = A$ and all A_{f_i} have the property, so does A . Define local properties of modules or algebras similarly using $(A, M) \mapsto (A_f, M_f)$.

Proposition. *Being the zero module, being a reduced ring, being a finitely generated module, and being a finitely generated algebra are local properties.*

Proof. The zero-module assertion was just proved. For reducedness use $(\sqrt{0})_f = \sqrt{0_{A_f}}$. If each M_{f_i} is generated by images of a finite set $S_i \subseteq M$, let M' be generated by their union. Then $(M/M')_{f_i} = 0$ for every i , so $M = M'$. The algebra case is analogous (exercise). $\square$

12 Semilattices and irreducibility

Definition. A *meet semilattice* is a poset C with a maximum 1_C and a greatest lower bound $x \wedge y$ for every pair. Maximum and meets are unique. They satisfy

$$1_C \wedge x = x, \quad x \wedge x = x, \quad x \wedge y = y \wedge x,$$

$$(x \wedge y) \wedge z = x \wedge (y \wedge z).$$

Conversely, an idempotent commutative monoid $(C, \wedge, 1)$ determines such an order by $x \leq y \iff x \wedge y = x$. An element x is *irreducible* if $x \neq 1_C$ and $x = y \wedge z$ implies $x = y$ or $x = z$. An irreducible component of x is a minimal irreducible element above x . An *irreducible decomposition* is a finite expression $x = x_1 \wedge \cdots \wedge x_n$ with all x_i irreducible. The empty meet is 1_C .

Example. On $\mathbb{N}$ with reverse divisibility, $a \leq b$ means $b \mid a$. The meet is lcm and the maximum is 1. The irreducibles are 0 and the positive prime powers. Prime factorization gives $a = p_1^{e_1} \wedge \cdots \wedge p_r^{e_r}$.

The closed subsets of a space, ordered by reverse inclusion, form a meet semilattice with meet union and maximum $\emptyset$. A space X is irreducible if it is nonempty and is not the union of two proper closed subsets. **Exercises.** A subset W is irreducible exactly when $\overline{W}$ is. For closed W , this agrees with semilattice irreducibility. Its components are the maximal irreducible subspaces of W . Submodules of M , ordered by inclusion, form a meet semilattice with meet intersection and maximum M .

13 Noetherian posets and spaces

Definition. A poset is *Noetherian* if every ascending sequence stabilizes. Equivalently, every nonempty subset has a maximal element. Every subposet of a Noetherian poset is Noetherian.

Equivalence. Without a maximal element one can successively choose strictly larger elements, contradicting stabilization. Conversely, a maximal member of an ascending sequence forces that sequence to stabilize. $\square$

Proposition. *Every element of a Noetherian meet semilattice has an irreducible decomposition.*

Proof. Otherwise choose a maximal element x without such a decomposition. It is neither the maximum nor irreducible, so $x = y \wedge z$ with $y, z > x$. Both have decompositions, whose combined meet decomposes x , a contradiction. $\square$

A space X is *Noetherian* if descending chains of closed sets stabilize.

Proposition. *Every subspace of a Noetherian space is Noetherian. Such a space has finitely many irreducible components $Z_1, \dots, Z_r$, and*

$$X = Z_1 \cup \dots \cup Z_r.$$

Every irreducible subset lies in some Z_i . The components are unique up to permutation.

Proof. For a subspace W , closure in X injects its closed-set poset into that of X , since $F = W \cap \overline{F}$ for closed $F \subseteq W$. The decomposition theorem gives a finite irreducible closed cover of X ; discard members contained in others. If Z is irreducible, its cover by the finitely many relatively closed sets $Z \cap Z_i$ forces $Z \subseteq Z_i$ for some i . Thus the remaining members are precisely the components. $\square$

14 Noetherian modules

Definition. An A -module is *Noetherian* if its submodule poset satisfies the ascending chain condition. A ring is Noetherian if it is so as a module over itself. **Exercises.**

1. In $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, the module M is Noetherian if and only if both M' and M'' are.
2. Finite direct sums of Noetherian modules are Noetherian.
3. M is Noetherian if and only if every submodule is finitely generated. Over a Noetherian ring, this is equivalent to M being finitely generated.

Proposition. *Localizations of Noetherian rings are Noetherian. Being a Noetherian module is a local property.*

Proof. Use the order-preserving injection from ideals of A_S to ideals of A given by contraction. For modules apply locality of finite generation to each submodule, using exactness of localization and the submodule correspondence. $\square$

Lemma. *A radical ideal is irreducible in the semilattice of radical ideals if and only if it is prime. This differs from irreducibility in the full ideal lattice.*

Proof. Suppose I is irreducible and $J_1 J_2 \subseteq I$. Replacing J_i by $J_i + I$ gives $I \subseteq J_i$, and

$$\sqrt{J_1} \cap \sqrt{J_2} = \sqrt{J_1 J_2} = I.$$

Thus $J_i \subseteq I$ for some i , proving primality. Conversely, if I is prime and $J_1 \cap J_2 = I$, then $J_1 J_2 \subseteq I$ forces one $J_i \subseteq I$, hence equality. $\square$

15 Noetherian spectra and decompositions

Proposition. *If A is Noetherian, then $\operatorname{Spec} A$ is Noetherian and A has only finitely many minimal primes.*

Proof. The closed-set poset of $\operatorname{Spec} A$, with reverse inclusion, is isomorphic to the poset of radical ideals, a subposet of the ideal lattice. Minimal primes correspond to irreducible components of $\operatorname{Spec} A$. $\square$

Every submodule of a Noetherian module has an irreducible decomposition, by the semilattice theorem.

Example. If a radical ideal I has finitely many minimal primes $\mathfrak{p}_1, \dots, \mathfrak{p}_n$, then $I = \mathfrak{p}_1 \cap \dots \cap \mathfrak{p}_n$ is an irreducible decomposition. In contrast, in $\mathbb{Q}[x, y]$ the ideal $(x, y)^2$ has infinitely many irreducible decompositions:

$$(x, y)^2 = ((x + ay)^2, y) \cap (x + ay, y^2) \quad (a \in \mathbb{Q}).$$

Remark. If M, N are Noetherian A -modules, so is $M \otimes_A N$: a finite generating set of N exhibits it as a quotient of a finite sum of copies of M . However, tensor products of Noetherian A -algebras need not be Noetherian rings; for example, consider $\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$.

16 Hilbert's basis theorem

Theorem. *If A is Noetherian, then $A[x]$ is Noetherian.*

Proof. Suppose an ideal $I \subseteq A[x]$ is not finitely generated. Choose $f_n \in I$ of least degree outside $(f_1, \dots, f_{n-1})$. Their degrees are nondecreasing. Let a_n be the leading coefficient of f_n . Since A is Noetherian, $(a_1, a_2, \dots) = (a_1, \dots, a_N)$ for some N . Write $a_{N+1} = \sum_{k=1}^N b_k a_k$. Then

$$g = f_{N+1} - \sum_{k=1}^N b_k x^{\deg f_{N+1} - \deg f_k} f_k$$

has smaller degree than f_{N+1} . Since $g \in I$, its degree forces $g \in (f_1, \dots, f_N)$, contradicting the choice of f_{N+1} . $\square$

Corollary. *Every finitely generated algebra over a Noetherian ring is Noetherian. If B is a finitely generated A -algebra and C is a Noetherian A -algebra, then $B \otimes_A C$ is Noetherian.*

Proof. Iterate the theorem and pass to quotients. For the last assertion, $B \otimes_A C$ is a finitely generated C -algebra. $\square$

17 Associated primes

Definition. For $M \in \text{Mod}_A$, define

$$\text{Supp}_A M = \{\mathfrak{p} \in \text{Spec } A : M_{\mathfrak{p}} \neq 0\},$$

$$\mathcal{A}_A(M) = \{\text{Ann}_A(m) : 0 \neq m \in M\}, \quad \text{Ass}_A M = \mathcal{A}_A(M) \cap \text{Spec } A.$$

The union of $\mathcal{A}_A(M)$ is the set of zerodivisors on M , and $\text{Ass}_A(A/\mathfrak{p}) = \{\mathfrak{p}\}$.

Lemma. 1. $\text{Supp } M \subseteq V(\text{Ann } M)$, with equality if M is finitely generated.

2. $\mathfrak{p} \in \text{Ass } M$ exactly when there is an injection $A/\mathfrak{p} \hookrightarrow M$.

3. $\text{Ass } M \subseteq \text{Supp } M$.

4. Every maximal member of $\mathcal{A}_A(M)$ is prime.

Proof. If $M_{\mathfrak{p}} \neq 0$, some m survives, so $\text{Ann}(m) \subseteq \mathfrak{p}$. If M has finite generators m_i and $M_{\mathfrak{p}} = 0$, choose $s_i \notin \mathfrak{p}$ killing m_i ; their product kills M , proving the converse in (1). Part (2) follows from $Am \simeq A/\text{Ann}(m)$, and localization of that injection gives (3). For (4), let $I = \text{Ann}(m)$ be maximal and suppose $ab \in I$, $a \notin I$. Since $am \neq 0$, maximality gives $\text{Ann}(am) = I$, hence $b \in I$. $\square$

For an exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$,

$$\text{Ass } M' \subseteq \text{Ass } M \subseteq \text{Ass } M' \cup \text{Ass } M''.$$

Indeed, an embedded $A/\mathfrak{p} \subseteq M$ either meets M' trivially and injects into M'' , or a nonzero element of the intersection generates another copy of $A/\mathfrak{p}$. In particular,

$$\text{Ass}(M' \oplus M'') = \text{Ass } M' \cup \text{Ass } M''.$$

18 Associated primes over Noetherian rings

Assume A is Noetherian. Every annihilator of a nonzero element is contained in a maximal such annihilator, so

$$\bigcup_{\mathfrak{p} \in \text{Ass } M} \mathfrak{p} = \{\text{zerodivisors on } M\}.$$

Thus $M \neq 0$ implies $\text{Ass } M \neq \emptyset$, and the localization map

$$M \longrightarrow \prod_{\mathfrak{p} \in \text{Ass } M} M_{\mathfrak{p}}$$

is injective: for $0 \neq m$, choose an associated prime containing $\text{Ann}(m)$.

Proposition. *A finitely generated M admits a prime filtration*

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M, \quad M_i/M_{i-1} \simeq A/\mathfrak{p}_i.$$

In particular, $\text{Ass } M$ is finite and contained in $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\} \subseteq \text{Supp } M$.

Proof. Choose a maximal submodule admitting such a filtration. If it is not all of M , a nonzero quotient has an associated prime, permitting one more step, a contradiction. $\square$

Proposition. *For a multiplicative S , contraction identifies*

$$\text{Ass}_{A_S}(M_S) = \text{Ass}_A(M) \cap (\text{Spec } A)_S.$$

Proof. Localizing $A/\mathfrak{p} \hookrightarrow M$ proves one inclusion. Conversely an injection $(A/\mathfrak{p})_S \hookrightarrow M_S$ is φ/s for a map $\varphi : A/\mathfrak{p} \rightarrow M$, because $A/\mathfrak{p}$ is finitely presented and Hom commutes with flat base change. If $\varphi(x) = 0$, its localized injection gives $x/1 = 0$. Since $\mathfrak{p} \cap S = \emptyset$, $A/\mathfrak{p} \rightarrow (A/\mathfrak{p})_S$ is injective; hence φ is injective. The argument uses the commuting square

$$\begin{array}{ccc} A/\mathfrak{p} & \xrightarrow{\varphi} & M \\ \downarrow & & \downarrow \\ (A/\mathfrak{p})_S & \xrightarrow{\varphi_S} & M_S. \end{array}$$

$\square$

19 Minimal support and primary modules

If A is Noetherian and M is finitely generated, every minimal prime in $\text{Supp } M$ belongs to $\text{Ass } M$. Indeed, localization at such a prime has nonempty associated-prime set and support consisting only of the maximal ideal. Consequently

$$\bigcap_{\mathfrak{p} \in \text{Ass } M} \mathfrak{p} = \sqrt{\text{Ann } M}.$$

If A is also reduced, $\text{Ass}_A A$ consists precisely of the minimal primes: embed A into the finite product of the $A/\mathfrak{p}$ for minimal primes $\mathfrak{p}$, using $\sqrt{0} = 0$. A module is $\mathfrak{p}$ -coprimary if $\text{Ass } M = \{\mathfrak{p}\}$. A submodule $N \subseteq M$ is $\mathfrak{p}$ -primary if M/N is $\mathfrak{p}$ -coprimary.

Proposition. *For a nonzero finitely generated module over a Noetherian ring, being coprimary is equivalent to every zerodivisor acting nilpotently on M . In that case $\mathfrak{p} = \sqrt{\text{Ann } M}$ is its associated prime.*

Proof. The union of associated primes is the set of zerodivisors, and their intersection is $\sqrt{\text{Ann } M}$. Equality of union and intersection is equivalent to having just one associated prime. $\square$

Examples. A proper ideal I is primary precisely when $ab \in I$ implies $b \in I$ or $a \in \sqrt{I}$. The module $\mathbb{Z}/n\mathbb{Z}$ is coprimary precisely when $n = 0$ or n is a positive prime power. For a minimal prime $\mathfrak{p}$ of $\text{Supp } M$, the module $M_{\mathfrak{p}}$ is $\mathfrak{p}$ -coprimary as an A -module. The kernel of $M \rightarrow M_{\mathfrak{p}}$ is the canonical $\mathfrak{p}$ -primary component of zero.

20 Existence of primary decompositions

Throughout, A is Noetherian and M is finitely generated.

Proposition. *Every irreducible proper submodule is primary. The intersection of two $\mathfrak{p}$ -primary submodules is $\mathfrak{p}$ -primary.*

Proof. Pass to the quotient to take the submodule to be zero. If $A/\mathfrak{p}_i \hookrightarrow M$ for $i = 1, 2$, irreducibility makes their images intersect nontrivially. A nonzero element of the intersection has annihilator both $\mathfrak{p}_1$ and $\mathfrak{p}_2$, proving uniqueness. For the second statement use

$$M/(N_1 \cap N_2) \hookrightarrow M/N_1 \oplus M/N_2,$$

whose nonzero left side has associated primes contained in $\{\mathfrak{p}\}$. □

A *primary decomposition* of $N \subseteq M$ is an expression $N = N_1 \cap \cdots \cap N_r$ with primary N_i . It is *irredundant* if deleting any component changes the intersection, and *minimal* if no primary decomposition uses fewer components.

Theorem. *Primary decompositions exist. If $\text{Ass}(M/N_i) = \{\mathfrak{p}_i\}$, then*

$$\text{Ass}(M/N) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\},$$

with equality for an irredundant decomposition. In a minimal decomposition the $\mathfrak{p}_i$ are distinct.

Proof. Irreducible decomposition gives existence. Pass to $N = 0$ and embed M in $\bigoplus_i M/N_i$ for the containment. For an irredundant decomposition, $M'_i = \bigcap_{j \neq i} N_j \neq 0$ injects into M/N_i , so its associated-prime set is $\{\mathfrak{p}_i\}$ and is contained in $\text{Ass } M$. Repeated primes can be combined by intersection, proving the last assertion. □

21 Uniqueness and localization of primary components

Let $N = \cap_i N_i$ be a minimal primary decomposition, with distinct associated primes $\mathfrak{p}_i$.

Theorem. *If $\mathfrak{p}_j$ is minimal in $\text{Supp}(M/N)$, then*

$$N_j = \ker(M \longrightarrow (M/N)_{\mathfrak{p}_j}).$$

Thus the components belonging to minimal primes are uniquely determined. For a multiplicative set S , localization retains precisely those components with $\mathfrak{p}_i \cap S = \emptyset$, and their intersection is a minimal primary decomposition of N_S .

Proof. Pass to $N = 0$. For $i \neq j$, minimality of $\mathfrak{p}_j$ implies $(M/N_i)_{\mathfrak{p}_j} = 0$, by the associated-prime localization formula. The injection into the sum of all primary quotients therefore localizes to the bottom injection in

$$\begin{array}{ccc} M & \longrightarrow & M/N_j \\ \downarrow & & \downarrow \\ M_{\mathfrak{p}_j} & \hookrightarrow & (M/N_j)_{\mathfrak{p}_j}. \end{array}$$

The right map is injective because every element outside $\mathfrak{p}_j$ is regular on M/N_j . Equality of the two kernels proves the formula. Finally, $(M/N_i)_S$ is zero if S meets $\mathfrak{p}_i$, and otherwise is $\mathfrak{p}_i A_S$ -coprimary. Exact localization preserves the intersection. Distinct surviving primes remain distinct. Each survives as an associated prime of $(M/N)_S$, so none of these components is redundant. $\square$

22 Cayley–Hamilton and the determinant trick

Theorem. *Suppose M is generated by n elements and $\varphi(M) \subseteq IM$. Then $p(\varphi) = 0$ for a monic polynomial*

$$p(t) = t^n + p_1 t^{n-1} + \cdots + p_n, \quad p_j \in I^j.$$

Proof. Choose generators m_i and a matrix $P = (a_{ij})$ with entries in I such that $\varphi(m_i) = \sum_j a_{ij} m_j$. View M as an $A[t]$ -module with t acting by φ . The column of generators satisfies $(tI_n - P)\mathbf{m} = 0$. Multiplication by the adjugate gives $\det(tI_n - P)M = 0$. Its coefficient of t^{n-j} lies in I^j . $\square$

Corollary. *A surjective endomorphism of a finitely generated module is an isomorphism. If $A \neq 0$ and $M \simeq A^n$, any n generators form a basis, and the rank n is well defined.*

Proof. Let t act by the surjective endomorphism α , so $tM = M$. Apply the theorem to the identity of M over $A[t]$ with ideal (t) . It yields $(1 - tq(t))M = 0$, so $q(\alpha)$ is an inverse to α . Apply this to a surjection $A^n \rightarrow A^n$ for the basis assertion. Rank invariance follows also by reduction modulo a maximal ideal. $\square$

The *Jacobson radical* is $J(A) = \bigcap_{\mathfrak{m} \in \text{Max } A} \mathfrak{m}$.

Corollary (Nakayama’s lemma). *If $I \subseteq J(A)$ and M is finitely generated, then $IM = M$ implies $M = 0$. A finite set generating M/IM lifts to a generating set of M .*

23 Finite and integral morphisms

Proof of Nakayama's lemma. The determinant trick applied to id_M gives $(1 - a)M = 0$ for some $a \in I$. Since $I \subseteq J(A)$, the element $1 - a$ belongs to no maximal ideal and is a unit. Hence $M = 0$. Apply this to the quotient by the proposed generators for the second assertion. $\square$

A homomorphism $A \rightarrow B$ is *finite* if B is finitely generated as an A -module. If B is finite over A and M is finite over B , then M is finite over A : products of finite generating sets generate. In particular, finite homomorphisms compose. Later we will see that their maps on spectra have finite fibers. An element $b \in B$ is *integral over A* if it satisfies a monic polynomial in $A[t]$. The algebra B is *integral* if all its elements are integral.

Proposition. *The following are equivalent:*

1. B is finite over A .
2. B is a finitely generated integral A -algebra.
3. B is generated as an algebra by finitely many integral elements.

Proof. For $(1) \Rightarrow (2)$, apply Cayley–Hamilton to multiplication by each $b \in B$ and evaluate the resulting endomorphism at 1. The implication $(2) \Rightarrow (3)$ is immediate. For $(3) \Rightarrow (1)$, induct on the generators. If b satisfies a monic polynomial p of degree d , then $A[b]$ is a quotient of $A[t]/(p)$, generated as a module by $1, t, \dots, t^{d-1}$. $\square$

24 Integral closure and normalization

Proposition. *Integral homomorphisms compose.*

Proof. For $A \rightarrow B \rightarrow C$ and $c \in C$, choose a monic equation for c with coefficients in B . Let B' be the A -algebra generated by those finitely many coefficients. Then B' is finite over A and $B'[c]$ is finite over B' , so c is integral over A . $\square$

Theorem. *The elements of an A -algebra B integral over A form an A -subalgebra. In particular, an algebra generated by integral elements is integral.*

Proof. If b_1, b_2 are integral, then $A[b_1, b_2]$ is finite over A . Hence $b_1 + b_2$, $b_1 b_2$, and $-b_1$ are integral. $\square$

This subalgebra is the *integral closure of A in B* . For a domain A , its *normalization* is its integral closure in $\text{Frac } A$. A domain equal to its normalization is *normal*, or integrally closed.

Example. For $K \subseteq \overline{\mathbb{Q}}$, the integral closure $\mathcal{O}_K$ of $\mathbb{Z}$ in K is its ring of integers. For a field k ,

$$k[x, y]/(y^2 - x^3) \simeq k[t^2, t^3] \subseteq k[t],$$

with $x \mapsto t^2$, $y \mapsto t^3$. Its normalization in $k(t)$ is $k[t]$.

25 Integrality and closed maps

A ring map $\varphi : A \rightarrow B$ is *closed* if $\varphi^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is closed. It is *universally closed* if every base change $C \rightarrow B \otimes_A C$ is closed. Quotient maps are universally closed, and closed or universally closed maps compose.

Lemma. *The fiber over $\mathfrak{p}$ is naturally homeomorphic to*

$$(\varphi^*)^{-1}(\mathfrak{p}) \simeq \operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p})).$$

For any ideal $I \subseteq B$,

$$\overline{\varphi^*(V_B(I))} = V_A(\varphi^{-1}(I)).$$

Proof. The fiber consists of primes containing $\mathfrak{p}B$ and disjoint from the image of $A \setminus \mathfrak{p}$, hence is the spectrum of $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$. For the second assertion reduce to an injection $A \hookrightarrow B$. If the image misses $D(f)$, the image of f is nilpotent in B , hence f is nilpotent in A . Thus the image is dense. $\square$

Proposition (Lying over and closedness). *If $A \hookrightarrow B$ is integral, then $\operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is surjective, and*

$$\varphi^*(V_B(I)) = V_A(I \cap A).$$

Proof. For lying over, localize to make $(A, \mathfrak{p}A_{\mathfrak{p}})$ local. If $1 \in \mathfrak{p}B$, an equation $1 = \sum a_i b_i$ with $a_i \in \mathfrak{p}$ lies in a finite subalgebra $B' = A[b_1, \dots, b_n]$. Then $\mathfrak{p}B' = B'$, contradicting Nakayama and $A \hookrightarrow B'$. Thus the fiber ring is nonzero and has a prime. Apply lying over to $A/(I \cap A) \hookrightarrow B/I$ for closedness. $\square$

26 Universal closedness

Theorem. *A ring homomorphism is integral if and only if it is universally closed.*

Proof. An integral map factors as a quotient followed by an integral inclusion, both closed. Base changes remain integral because images of integral generators are integral, proving one direction. Conversely suppose $A \rightarrow B$ is universally closed and fix $b \in B$. Consider

$$\psi : A[x] \longrightarrow B[x] \longrightarrow B_b, \quad x \longmapsto b^{-1}.$$

The second map is surjective: $B_b = B[x]/(bx - 1)$. Thus ψ is universally closed. Let $J = \ker \psi$. The map $\text{Spec } B_b \rightarrow V(J)$ is both dense and closed, hence surjective. Its fiber over $V(x) \cap V(J)$ is empty since x maps to a unit. Therefore $A[x]/(J + (x)) = 0$. Choose $f \in J$ with $f(0) = 1$, say $f(x) = 1 + a_1x + \cdots + a_nx^n$. The equality $f(b^{-1}) = 0$ in B_b gives, after clearing denominators, for some $r \geq 0$,

$$b^{n+r} + a_1b^{n+r-1} + \cdots + a_nb^r = 0.$$

This is a monic equation for b over A . □

27 Going up, going down, and dimension

A map has *lying over* if its map on spectra is surjective. It has *going up* if, given $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ and $\mathfrak{q}_1$ over $\mathfrak{p}_1$, one can find $\mathfrak{q}_2 \supseteq \mathfrak{q}_1$ over $\mathfrak{p}_2$. It has *going down* if a prime $\mathfrak{q}_2$ over $\mathfrak{p}_2$ can instead be extended down to $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ over $\mathfrak{p}_1$. Integral inclusions have lying over. Integral maps have going up, by applying lying over to the induced inclusion after quotienting by $\mathfrak{q}_1$ and its contraction.

Proposition (Incomparability). *If $A \subseteq B$ is integral and $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ have the same contraction, then $\mathfrak{q}_1 = \mathfrak{q}_2$.*

Proof. For domains $A \subseteq B$ with algebraic fraction-field extension, every nonzero ideal of B meets A nontrivially: for $0 \neq b$ in it, clear denominators in an algebraic equation for b with nonzero constant term. That constant belongs to the ideal. Apply this after passing to $A/(\mathfrak{q}_1 \cap A) \subseteq B/\mathfrak{q}_1$. $\square$

The *Krull dimension* is

$$\dim A = \sup\{n : \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n \text{ in } \operatorname{Spec} A\}.$$

Corollary. *An integral extension $A \subseteq B$ satisfies $\dim A = \dim B$.*

Proof. Going up lifts every finite prime chain in A ; incomparability makes contraction preserve strictness of every prime chain in B . $\square$

Examples. $\dim \mathbb{Z} = 1$, and $\dim \mathcal{O}_K = 1$ for $K \subseteq \overline{\mathbb{Q}}$.

28 Jacobson rings

A ring A is *Jacobson* if every prime is the intersection of the maximal ideals containing it.

Lemma. *If A is a Jacobson domain and A_s is a field, then A is a field.*

Proof. The only prime avoiding s is (0) . If (0) is not maximal, every maximal ideal contains s , so $s \in J(A) = 0$, impossible since A_s is a field. $\square$

Examples. $\mathbb{Z}$ and $k[x]$ for a field k are Jacobson. Their nonzero primes are maximal, and the intersection of maximal ideals is zero: a nonzero element has only finitely many irreducible factors, whereas there are infinitely many primes or irreducible polynomials.

Theorem (Nullstellensatz for Jacobson rings). *If A is Jacobson, then $A[x]$ is Jacobson. Every maximal ideal $\mathfrak{n} \subseteq A[x]$ contracts to a maximal ideal $\mathfrak{m} \subseteq A$, and $\kappa(\mathfrak{n})$ is finite over $\kappa(\mathfrak{m})$.*

Proof of the maximal-ideal assertion. Pass to $A/(\mathfrak{n} \cap A)$, a Jacobson domain embedded in the field $K = A[x]/\mathfrak{n}$. The image of x must be algebraic over $\text{Frac } A$: otherwise $A[x]$ itself would be a field, which it is not. Inverting a nonzero leading coefficient $t \in A$ makes that image integral, so K is finite over A_t . A subring over which a field is integral is itself a field: a monic equation for a^{-1} shows a^{-1} belongs to the subring. Hence A_t , and by the lemma A , are fields. The residue extension is therefore finite. $\square$

29 The Nullstellensatz and affine algebraic sets

Completion of the Jacobson-ring proof. It suffices to show that if $P \in \operatorname{Spec} A[x]$ and $(A[x]/P)_s$ is a field for some nonzero image of s , then P is maximal. Put $B = A[x]/P$ and replace A by $A/(P \cap A)$. If the image of x is transcendental over $\operatorname{Frac} A$, further localization gives a field of the form $(\operatorname{Frac}(A)[x])_s$, impossible by the preceding lemma applied to the Jacobson domain $\operatorname{Frac}(A)[x]$. Otherwise B_s is finite algebraic over $\operatorname{Frac} A$. Choose $0 \neq t \in A$ making both the image of x and s^{-1} integral over A_t . The field B_s is finite over A_t , so A_t is a field. The lemma makes A a field; then the algebraic domain $A[x]/P$ is a field, as required. $\square$

Thus every finitely generated algebra over $\mathbb{Z}$ or a field is Jacobson, by iteration and passage to quotients. For an algebraically closed field k , evaluation gives bijections

$$k^n \simeq \operatorname{Hom}_{\mathbf{Alg}_k}(k[x_1, \dots, x_n], k) \simeq \operatorname{Max} k[x_1, \dots, x_n],$$

with $(a_i) \mapsto (x_1 - a_1, \dots, x_n - a_n)$. Indeed the residue extension at a maximal ideal is finite over k , hence equal to k . Consequently radical ideals correspond to affine algebraic sets: $I(V(I)) = \sqrt{I}$. This follows because a radical ideal in a Jacobson ring is the intersection of the maximal ideals containing it. The coordinate-ring functor gives a *contravariant* equivalence between affine algebraic sets over k with regular maps and reduced finitely generated k -algebras:

$$V \longmapsto k[V] = k[x_1, \dots, x_n]/I(V), \quad f : V \rightarrow W \longmapsto f^* : k[W] \rightarrow k[V].$$

Full faithfulness is an exercise using coordinate functions. Essential surjectivity follows by presenting a reduced algebra as a polynomial ring modulo a radical ideal.

30 Graded rings and filtrations

Definition. A $\mathbb{Z}$ -graded ring is $A = \bigoplus_{d \in \mathbb{Z}} A_d$ with $A_d A_e \subseteq A_{d+e}$. A graded module $M = \bigoplus_d M_d$ satisfies $A_d M_e \subseteq M_{d+e}$. An $\mathbb{N}$ -graded ring has $A_d = 0$ for $d < 0$. The ring A_0 acts on every graded piece and on A . Fix an ideal $I \subseteq A$. An I -filtration is

$$M = F^0 M \supseteq F^1 M \supseteq \cdots, \quad IF^n M \subseteq F^{n+1} M.$$

It is *stable* if equality holds for all sufficiently large n . The Rees algebra (or blowup algebra) and Rees module are

$$\mathcal{R}_I(A) = \bigoplus_{n \geq 0} I^n t^n \subseteq A[t], \quad \mathcal{R}_F(M) = \bigoplus_{n \geq 0} F^n M t^n.$$

Proposition. *If every $F^n M$ is finitely generated, the filtration is stable exactly when $\mathcal{R}_F(M)$ is finite over $\mathcal{R}_I(A)$.*

Proof. If homogeneous generators have degrees at most c , then $F^{c+j} M = I^j F^c M$ for $j \geq 0$. Conversely, this identity means generators of $F^0 M, \dots, F^c M$ generate the Rees module. $\square$

31 Artin–Rees and Krull intersection

Lemma (Artin–Rees). *Let A be Noetherian, M finite, and $N \subseteq M$. If F is a stable I -filtration on M , the induced filtration $F^n M \cap N$ on N is stable. In particular, for some c and every $n \geq c$,*

$$I^n M \cap N = I^{n-c}(I^c M \cap N).$$

Proof. Since I is finite, $\mathcal{R}_I(A)$ is a finitely generated A -algebra, hence Noetherian. The Rees module of the induced filtration is a submodule of the finite module $\mathcal{R}_F(M)$ and is therefore finite. Apply the previous proposition. $\square$

Corollary (Krull intersection). *For a finite module M over a Noetherian ring, some $a \in I$ satisfies*

$$(1 - a) \bigcap_{n \geq 0} I^n M = 0.$$

If A is a domain or a local ring and $I \neq A$, then $\bigcap_n I^n = 0$. More generally, if $I \subseteq J(A)$, then $\bigcap_n I^n M = 0$ for every finite M .

Proof. Set $N = \bigcap_n I^n M$. Artin–Rees gives $N = IN$. The determinant trick gives $(1 - a)N = 0$. In a domain with $a \in I \neq A$, $1 - a \neq 0$; in the local or Jacobson-radical case it is a unit. $\square$

32 Associated graded rings and initial forms

The associated graded ring and module are

$$\mathrm{gr}_I A = \bigoplus_{n \geq 0} I^n / I^{n+1}, \quad \mathrm{gr}_F M = \bigoplus_{n \geq 0} F^n M / F^{n+1} M.$$

The latter is a graded module over the former. The correct Rees description is

$$\mathrm{gr}_I A \simeq \mathcal{R}_I(A) / I\mathcal{R}_I(A).$$

If F is stable and its pieces are finite, then $\mathrm{gr}_F M$ is finite over $\mathrm{gr}_I A$.

Proof. The Rees module is finite, and its quotient by $I\mathcal{R}_F(M)$ is finite over $\mathrm{gr}_I A$. The natural surjection from this quotient onto $\mathrm{gr}_F M$ proves the assertion, since $IF^n M \subseteq F^{n+1} M$. $\square$

For a Noetherian local ring $(A, \mathfrak{m}, k)$, $\mathrm{gr}_{\mathfrak{m}} A$ is a finitely generated graded k -algebra. For $m \in M$, its *initial form* is its image in $F^n M / F^{n+1} M$, where n is the largest integer with $m \in F^n M$. If m lies in every piece, set its initial form to zero.

Proposition. *If A is Noetherian local, $I \neq A$, and $\mathrm{gr}_I A$ is a domain, then A is a domain.*

Proof. Krull intersection makes the filtration separated, so every nonzero element has a nonzero initial form. If $fg = 0$ with $f, g \neq 0$, the product of their initial forms is zero, contradicting that the graded ring is a domain. $\square$

33 Complexes and resolutions

Definition. A chain complex of A -modules has differentials lowering degree:

$$\cdots \rightarrow M_{n+1} \xrightarrow{d_{n+1}} M_n \xrightarrow{d_n} M_{n-1} \rightarrow \cdots, \quad d_n d_{n+1} = 0.$$

A chain map satisfies $\varphi_{n-1} d_n = d_n \varphi_n$. Modules embed as complexes concentrated in degree zero. The homology functor is

$$H_n(M) = \ker d_n / \operatorname{Im} d_{n+1}, \quad H_n(\varphi)[m] = [\varphi_n(m)].$$

A map is a *quasi-isomorphism* if it induces isomorphisms on all homology groups. In cohomological notation, $M^n = M_{-n}$, with differential raising degree, and $H^n(M^\bullet) = H_{-n}(M_\bullet)$. Two maps $\varphi_0, \varphi_1 : M \rightarrow N$ are homotopic if maps $h_n : M_n \rightarrow N_{n+1}$ satisfy

$$\varphi_1 - \varphi_0 = d_N h + h d_M.$$

Exercise. Homotopic maps induce the same homology maps. A *left resolution* of a module M is a quasi-isomorphism $P_\bullet \rightarrow M$ with $P_n = 0$ for $n < 0$. A *right resolution* is a quasi-isomorphism $M \rightarrow I^\bullet$ with $I^n = 0$ for $n < 0$. A module P is projective if $\operatorname{Hom}_A(P, -)$ is exact, equivalently if it is a direct summand of a free module. A module I is injective if $\operatorname{Hom}_A(-, I)$ is exact.

34 Defining Tor

Definition. A resolution is free, projective, or flat when all its terms are so. Every module has a free resolution, obtained by successively choosing free surjections onto kernels.

Theorem (Comparison and horseshoe lemmas). *A map $f : M \rightarrow N$ lifts from a projective resolution $P_\bullet \rightarrow M$ to any left resolution $Q_\bullet \rightarrow N$, uniquely up to chain homotopy:*

$$\begin{array}{ccc} P_\bullet & \xrightarrow{\varepsilon_M} & M \\ \tilde{f} \downarrow & & \downarrow f \\ Q_\bullet & \xrightarrow{\varepsilon_N} & N. \end{array}$$

Given $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ and projective resolutions P', P'' , there is a projective resolution P of M with $P_n = P'_n \oplus P''_n$ and a commuting diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & P' & \longrightarrow & P & \longrightarrow & P'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' \longrightarrow 0. \end{array}$$

For these proofs, see Weibel, Chapter 1. Fix free resolutions $F_\bullet(M) \rightarrow M$. Define

$$\mathrm{Tor}_n^A(M, N) = H_n(F_\bullet(M) \otimes_A N).$$

A map $M \rightarrow M'$ acts through a comparison lift and its induced homology map. Homotopy invariance makes this independent of the lift. Comparison also gives canonical compatible isomorphisms between computations from different projective resolutions. In fact any flat resolution computes Tor.

35 Basic properties of Tor

For $M, N \in \text{Mod}_A$,

1. $\text{Tor}_0^A(M, N) \simeq M \otimes_A N$.
2. Higher Tor vanishes if M is free or N is flat.
3. If A is Noetherian and M, N are finite, all $\text{Tor}_n^A(M, N)$ are finite.
4. For a flat A -algebra B ,

$$B \otimes_A \text{Tor}_n^A(M, N) \simeq \text{Tor}_n^B(M_B, N_B).$$

Proof. Right exactness of tensor gives (1). Exact tensoring with a flat module commutes with homology, giving (2). Over a Noetherian ring, a finite module admits a resolution by finite free modules, giving (3). For (4), tensor such a free resolution (not necessarily finite) with B ; flatness keeps it exact and commutes with its homology. $\square$

An exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ gives a natural long exact sequence

$$\begin{aligned} \cdots \rightarrow \text{Tor}_n(M', N) \rightarrow \text{Tor}_n(M, N) \rightarrow \text{Tor}_n(M'', N) \\ \rightarrow \text{Tor}_{n-1}(M', N) \rightarrow \cdots \\ \rightarrow \text{Tor}_1(M'', N) \rightarrow M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0. \end{aligned}$$

Use the horseshoe lemma: the resulting degreewise split exact sequence of resolutions remains exact after tensoring. An exact sequence in the second argument similarly gives a long exact sequence, because each free term in a resolution of M tensors exactly. **Exercise.** If $P_\bullet \rightarrow N$ is projective, prove

$$\text{Tor}_n^A(M, N) \simeq H_n(M \otimes_A P_\bullet) \simeq \text{Tor}_n^A(N, M).$$

36 Flatness and Tor

Definition. A category is *filtered* if it is nonempty, any two objects map to a common object, and any two parallel arrows become equal after a further arrow. In module categories, filtered colimits commute with finite limits and are exact.

Lemma. *M is flat if tensoring preserves every inclusion $P_0 \subseteq P$ with P finitely generated.*

Proof. Write N as the filtered union of its finitely generated submodules P . Then $N_0 = \operatorname{colim}_P (P \cap N_0)$ for any $N_0 \subseteq N$. Tensor commutes with these colimits, and exactness of filtered colimits preserves the assumed injections. $\square$

Proposition. *For any ideal I ,*

$$\operatorname{Tor}_1^A(A/I, M) = 0 \iff I \otimes_A M \longrightarrow IM \text{ is an isomorphism.}$$

Moreover M is flat if and only if this vanishing holds for every finitely generated ideal I .

Proof. Apply the Tor sequence to $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$:

$$0 \rightarrow \operatorname{Tor}_1^A(A/I, M) \rightarrow I \otimes_A M \rightarrow M \rightarrow M/IM \rightarrow 0.$$

If the condition holds for finite ideals, express any ideal as their filtered union to obtain injectivity for every ideal. For a finite module N and $N_0 \subseteq N$, filter N/N_0 by successively adjoining a finite set of generators. Its factors are cyclic modules A/I_j . The long exact sequences show $\operatorname{Tor}_1^A(N/N_0, M) = 0$. Thus $N_0 \otimes M \rightarrow N \otimes M$ is injective; apply the lemma. $\square$

37 The equational criterion for flatness

If N has a presentation $A^{(J)} \xrightarrow{(a_{ij})} A^{(I)} \rightarrow N \rightarrow 0$ with basis vectors mapping to n_i , then

$$\sum_i m_i \otimes n_i = 0 \iff m_i = \sum_j a_{ij} m'_j \text{ for some } (m'_j) \in M^{(J)}.$$

Each column $(a_{ij})_i$ is finitely supported and satisfies $\sum_i a_{ij} n_i = 0$. This is simply right exactness of tensor applied to the presentation.

Theorem (Equational criterion). *M is flat if and only if every finite relation $\sum_{i=1}^r a_i m_i = 0$ admits elements $m'_1, \dots, m'_s \in M$ and coefficients $b_{ij} \in A$ such that*

$$m_i = \sum_j b_{ij} m'_j, \quad \sum_i a_i b_{ij} = 0 \text{ for every } j.$$

Proof. Apply the presentation observation to the ideal $(a_1, \dots, a_r)$ and use the ideal criterion for flatness. $\square$

Equivalently, whenever $\beta : F \rightarrow M$ has F finite free and $K \subseteq \ker \beta$ is finitely generated, there is a factorization

$$\begin{array}{ccc} F & \xrightarrow{\beta} & M \\ & \searrow \gamma & \nearrow \delta \\ & G & \end{array}$$

with G finite free and $K \subseteq \ker \gamma$. The case of a principal K restates the theorem; kill finitely many generators successively for the general case.

Corollary. *A finitely presented module is flat if and only if it is projective.*

Proof. For a finite presentation $K \rightarrow F \rightarrow M \rightarrow 0$, the factorization with $\gamma(K) = 0$ descends to $M \rightarrow G$. Its composite with $G \rightarrow M$ is the identity, making M a direct summand of a free module. $\square$

38 The local criterion for flatness

Theorem. *Let $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ be a local homomorphism of Noetherian local rings and let M be a finite B -module. Then*

$$M \text{ is flat over } A \iff \mathrm{Tor}_1^A(A/\mathfrak{m}, M) = 0.$$

Proof. Assume the vanishing. Induction on length gives $\mathrm{Tor}_1^A(L, M) = 0$ for every finite-length A -module L , in particular $A/(\mathfrak{m}^n + I)$. For an ideal $I \subseteq A$, set $\varphi : I \otimes_A M \rightarrow M$. Tensoring the sequence

$$0 \rightarrow I/(I \cap \mathfrak{m}^n) \rightarrow A/\mathfrak{m}^n \rightarrow A/(I + \mathfrak{m}^n) \rightarrow 0$$

gives an injection on the left. Hence in the square

$$\begin{array}{ccc} I \otimes_A M & \longrightarrow & (I/(I \cap \mathfrak{m}^n)) \otimes_A M \\ \varphi \downarrow & & \downarrow \\ M & \longrightarrow & M/\mathfrak{m}^n M \end{array}$$

every element of $\ker \varphi$ comes from $(I \cap \mathfrak{m}^n) \otimes_A M$. Artin–Rees gives $I \cap \mathfrak{m}^{n+c} \subseteq \mathfrak{m}^n I$ for some c . Therefore

$$\ker \varphi \subseteq \bigcap_n \mathfrak{m}^n (I \otimes_A M) = 0.$$

The last equality is Krull intersection over B : $I \otimes_A M$ is finite over B and $\mathfrak{m}B \subseteq \mathfrak{n}$. The ideal criterion now proves flatness. The reverse implication is immediate. $\square$

39 Flatness modulo a regular element

Lemma. *If $x \in A$ is both A -regular and M -regular, then for every $A/(x)$ -module N ,*

$$\mathrm{Tor}_n^{A/(x)}(N, M/xM) \simeq \mathrm{Tor}_n^A(N, M).$$

Proof. For a free resolution $F_\bullet \rightarrow M$, the complex $F_\bullet/xF_\bullet$ is a free resolution of M/xM . Indeed its homology is $\mathrm{Tor}_*^A(A/(x), M)$, computed using

$$0 \rightarrow A \xrightarrow{x} A \rightarrow A/(x) \rightarrow 0.$$

Regularity on M makes positive homology vanish. Now

$$N \otimes_A F_\bullet \simeq N \otimes_{A/(x)} (F_\bullet/xF_\bullet),$$

which proves the result. □

Corollary. *Under the hypotheses of the local criterion, suppose $x \in \mathfrak{m}$ is A -regular and M -regular. Then M is flat over A if and only if M/xM is flat over $A/(x)$.*

Proof. One direction is base change. For the other, the lemma identifies $\mathrm{Tor}_1^A(A/\mathfrak{m}, M)$ with $\mathrm{Tor}_1^{A/(x)}(A/\mathfrak{m}, M/xM)$, which vanishes. Apply the local criterion. □

40 Filtered rings and modules

Definition. A filtered ring (A, F) has a descending filtration by ideals

$$A = F^0 A \supseteq F^1 A \supseteq \cdots, \quad F^i A F^j A \subseteq F^{i+j} A.$$

A filtered morphism preserves all filtration pieces. The I -adic filtration is $F^n A = I^n$. Rings embed with the discrete filtration $F^1 A = 0$. If $I \subseteq F^1 A$, the identity defines a filtered map $(A, I) \rightarrow (A, F)$. A filtered module has $M = F^0 M \supseteq F^1 M \supseteq \cdots$ with $F^i A F^j M \subseteq F^{i+j} M$. An A -linear filtered map preserves these pieces. The I -adic filtration on a module is $I^n M$. Rees and associated-graded constructions are functorial:

$$\mathcal{R}_F(A) = \bigoplus_{n \geq 0} F^n A t^n, \quad \text{gr}_F A = \bigoplus_{n \geq 0} F^n A / F^{n+1} A,$$

and similarly for modules. A sequence (m_j) is *Cauchy* if for each n there is j_0 such that $m_j - m_k \in F^n M$ for all $j, k \geq j_0$. It is *null* if for each n , eventually $m_j \in F^n M$. It converges to m if $(m_j - m)$ is null. The module is *Hausdorff*, or separated, if $\bigcap_n F^n M = 0$. In these notes *complete* means that every Cauchy sequence converges uniquely, so includes separatedness.

41 Completion and its canonical filtration

Definition. Define

$$\widehat{M} = \varprojlim_n M/F^n M, \quad \widehat{F}^n M = \ker(\widehat{M} \rightarrow M/F^n M).$$

The natural map $c_M : M \rightarrow \widehat{M}$ sends m to its compatible residue classes. The completion $\widehat{A}$ is a filtered ring, and $\widehat{M}$ is a filtered $\widehat{A}$ -module.

Proposition. $\widehat{M}$ is complete for its canonical filtration, and

$$\widehat{M}/\widehat{F}^n M \simeq M/F^n M, \quad \text{gr}_{\widehat{F}} \widehat{M} \simeq \text{gr}_F M.$$

It also identifies with Cauchy sequences modulo null sequences.

Proof. Projection onto $M/F^n M$ is surjective, since any residue class has a representative in M . A Cauchy sequence determines an eventually constant class in each quotient. Conversely, choose representatives m_n of a compatible family; they form a Cauchy sequence. Null sequences are exactly those giving zero in every quotient. Taking these quotients again proves completeness and the graded identification. $\square$

Any filtered map $M \rightarrow N$ with N complete extends uniquely to a filtered map $\widehat{M} \rightarrow N$, by taking limits of images of Cauchy sequences. Functoriality is expressed by

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ c_M \downarrow & & \downarrow c_N \\ \widehat{M} & \xrightarrow{\widehat{\varphi}} & \widehat{N}. \end{array}$$

Remark (Keep the two filtrations distinct). For $\widehat{A} = \varprojlim A/I^n$, write $K_n = \ker(\widehat{A} \rightarrow A/I^n)$. Without additional hypotheses one only has

$$I^n \widehat{A} \subseteq (K_1)^n \subseteq K_n.$$

Do not identify these ideals merely from the definition of completion. The canonical filtration always makes $\widehat{A}$ complete; this alone is not a proof of completeness for powers of $I\widehat{A}$.

42 Lifting from associated graded modules

Proposition. *For a filtered map $\varphi : M \rightarrow N$, surjectivity of $\text{gr } \varphi$ implies surjectivity of $\widehat{\varphi}$. Injectivity of $\text{gr } \varphi$ implies injectivity of $\widehat{\varphi}$.*

Proof. Replace the modules by their completions, whose graded modules are unchanged. Given $y \in \widehat{N}$, lift its degree-zero class, then the degree-one error, and so on. This gives $x_n \in \widehat{F^n M}$ with

$$y - \varphi \left(\sum_{j=0}^{n-1} x_j \right) \in \widehat{F^n N}.$$

The sum converges in $\widehat{M}$, and its image is y . For injectivity, any nonzero element has a first nonzero graded class by separatedness; its image remains nonzero under an injective graded map. $\square$

Remark (A completeness requirement). This proposition concerns the map on *completions*. Surjectivity on associated graded modules need not make the original map surjective if its source is not complete. In particular, one cannot simply assert that a finitely generated ideal equals another ideal because their initial forms generate the same graded ideal; the needed convergent coefficient expansions must be justified.

Theorem. *If A is Noetherian, its I -adic completion is Noetherian and*

$$K_n = I^n \widehat{A} = (I \widehat{A})^n.$$

Thus $\widehat{A}$ is complete for the $I \widehat{A}$ -adic topology, with the same associated graded ring as A .

Proof strategy. The exactness and finite-module comparison proved next identify the kernel K_n with the image of $\widehat{I^n} \simeq I^n \otimes_A \widehat{A}$, giving the equality. Since $\text{gr}_I A$ is Noetherian, lift finite homogeneous generators of the initial ideal of an ideal $J \subseteq \widehat{A}$. Successively subtract their multiples to increase the order of a given element of J . The coefficients converge in $\widehat{A}$, expressing that element in the ideal of the finitely many lifts. Hence J is finitely generated. $\square$

43 Exactness of completion for finite modules

Throughout this page, A is Noetherian, $I \subseteq A$, and every module being completed is finitely generated. Completion is with respect to powers of I .

Proposition. *An exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ induces*

$$0 \rightarrow \widehat{M'} \rightarrow \widehat{M} \rightarrow \widehat{M''} \rightarrow 0.$$

Proof. For each n there is an exact sequence

$$0 \rightarrow M'/(M' \cap I^n M) \rightarrow M/I^n M \rightarrow M''/I^n M'' \rightarrow 0.$$

The transition maps in the left system are surjective. A compatible lift can therefore be constructed successively, correcting each lift by a left-hand term; thus taking inverse limits preserves surjectivity here. Left exactness gives the other two places. Artin–Rees makes the induced filtration $M' \cap I^n M$ cofinal with $I^n M'$. Hence its inverse limit is $\widehat{M'}$. $\square$

Theorem. *The natural map $\widehat{A} \otimes_A M \rightarrow \widehat{M}$ is an isomorphism for finite M . The ring $\widehat{A}$ is flat over A .*

Proof. Choose a finite presentation $A^r \rightarrow A^s \rightarrow M \rightarrow 0$. Completion of finite free modules is $\widehat{A}^r$, and the comparison diagram is

$$\begin{array}{ccccccc} \widehat{A} \otimes A^r & \longrightarrow & \widehat{A} \otimes A^s & \longrightarrow & \widehat{A} \otimes M & \longrightarrow & 0 \\ \downarrow \sim & & \downarrow \sim & & \downarrow & & \\ \widehat{A}^r & \longrightarrow & \widehat{A}^s & \longrightarrow & \widehat{M} & \longrightarrow & 0. \end{array}$$

Both rows are right exact, proving the comparison. For every ideal $J \subseteq A$, exactness gives $\widehat{A} \otimes_A J \simeq \widehat{J} \hookrightarrow \widehat{A}$. The ideal criterion proves flatness. $\square$

44 Power series and substitution

Let B be an A -algebra complete and separated for an ideal $\mathfrak{n}$, and choose $f_1, \dots, f_r \in \mathfrak{n}$. Evaluation extends uniquely as a *continuous* A -algebra homomorphism

$$\hat{\varphi} : A[[x_1, \dots, x_r]] \rightarrow B, \quad g \mapsto g(f_1, \dots, f_r).$$

If $A \rightarrow B/\mathfrak{n}$ is surjective and $(f_1, \dots, f_r) = \mathfrak{n}$, this map is surjective: its associated graded map is surjective. If its graded map is injective, the completed map is injective as well.

Corollary. *For $f \in A[[x]]$ with $f(0) = 0$, substitution $x \mapsto f$ is an automorphism if and only if $f'(0) \in A^\times$.*

Proof. An automorphism induces an automorphism of $(x)/(x^2)$, so its linear coefficient must be a unit. Conversely write $f = x(a + xh)$ with $a \in A^\times$. The series $a + xh$ has inverse

$$a^{-1} \sum_{j \geq 0} (-a^{-1} xh)^j.$$

Thus $(f) = (x)$, and the graded map is the automorphism sending x to ax . Apply the preceding lifting criterion. $\square$

45 Hensel's lemma

Theorem. *Let A be complete and separated for an ideal $\mathfrak{m}$. If $f \in A[x]$, $a \in A$, and*

$$f(a) \in f'(a)^2 \mathfrak{m},$$

then there exists $b \in A$ with $f(b) = 0$ and $b - a \in f'(a)\mathfrak{m}$. If $f'(a)$ is a nonzerodivisor, this root is unique among elements in that congruence class.

Proof. Translate so that $a = 0$ and put $e = f'(0)$. Write $f(0) = e^2 c$ with $c \in \mathfrak{m}$. Taylor expansion gives

$$f(ex) = e^2(c + x + x^2 h(x)).$$

The series $g(x) = x + x^2 h(x)$ has a formal inverse $p(x)$ by the substitution criterion. Since $c \in \mathfrak{m}$, $x_0 = p(-c)$ converges in $\mathfrak{m}$ and $b = ex_0$ is a root. If e is regular, any root $b = ex$ with $x \in \mathfrak{m}$ forces $g(x) = -c$, whose unique solution is $p(-c)$. $\square$

Example. For a prime p , the p -adic integers are $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n \mathbb{Z}$. If the reduction of $f \in \mathbb{Z}_p[x]$ has a simple root $a \in \mathbb{F}_p$, that root lifts uniquely to $\hat{a} \in \mathbb{Z}_p$ with $f(\hat{a}) = 0$. Applying this to $x^{p-1} - 1$ gives all $(p-1)$ st roots of unity and

$$\mathbb{Z}_p^\times \simeq \mu_{p-1} \times (1 + p\mathbb{Z}_p).$$

46 Cohen's structure theorem

Theorem. *If $(A, \mathfrak{m}, k)$ is a complete Noetherian local ring containing a field, then*

$$A \simeq k[[x_1, \dots, x_r]]/J$$

for some r and ideal J , after choosing a coefficient field $k \hookrightarrow A$ splitting the residue map.

Proof in characteristic zero. Once a section $k \hookrightarrow A$ exists, choose generators $f_1, \dots, f_r$ of $\mathfrak{m}$. Power-series substitution gives the required surjection $k[[x_1, \dots, x_r]] \rightarrow A$. Suppose $F \subseteq A$ is a field of characteristic zero. Its map into k is injective. Choose a transcendence basis T of k/F and representatives $\tilde{T} \subseteq A$. Polynomial algebraic independence and invertibility of elements with nonzero residue give an embedded field $F(\tilde{T}) \subseteq A$ mapping isomorphically onto $F(T)$. By maximality (or Zorn's lemma), choose a largest subfield $L \subseteq k$ over $F(T)$ equipped with a compatible section $\tilde{L} \subseteq A$:

$$\begin{array}{ccc} \tilde{L} & \hookrightarrow & A \\ \sim \downarrow & & \downarrow \\ L & \hookrightarrow & k. \end{array}$$

If $a \in k$ is algebraic over L , its minimal polynomial is separable. Lift its coefficients to $\tilde{L}$ and choose a lift of a . The derivative has nonzero residue, so Hensel's lemma lifts a to a root in A , extending the section to $L(a)$. Maximality forces $L = k$. $\square$

Remark. The theorem includes positive equal characteristic, but the proof above establishes only characteristic zero. The positive-characteristic coefficient-field argument is not supplied here.

47 Artinian rings and modules

Definition. A poset is *Artinian* if every descending chain stabilizes. A module is Artinian if its submodule poset is; a ring is Artinian as a module over itself. **Exercises.** A vector space is Noetherian if and only if it is Artinian if and only if it is finite dimensional. In a short exact sequence, the middle module is Artinian exactly when both end modules are.

Lemma. *If $\mathfrak{m}_1 \cdots \mathfrak{m}_r = 0$ for finitely many maximal ideals, with repetitions allowed, then A is Noetherian if and only if it is Artinian.*

Proof. Filter A by the successive products $I_j = \mathfrak{m}_1 \cdots \mathfrak{m}_j$. The quotient I_{j-1}/I_j is a vector space over $A/\mathfrak{m}_j$. Apply the exercises to this finite filtration. $\square$

Theorem. *A nonzero ring is Artinian if and only if it is Noetherian of dimension zero.*

Proof. If A is Noetherian of dimension zero, a primary decomposition of zero has finitely many maximal radicals $\mathfrak{m}_i$. Each primary component contains a power of its radical, so a finite product of maximal ideals is zero. Apply the lemma. Conversely, an Artinian domain is a field: stabilization of (x^n) for $x \neq 0$ gives $x^n = x^{n+1}y$, hence $xy = 1$. Thus all primes of an Artinian ring are maximal. There are finitely many maximal ideals: otherwise successive intersections of distinct maximal ideals form a strictly descending chain, by comaximality. Put $R = \sqrt{0} = \bigcap_i \mathfrak{m}_i = \prod_i \mathfrak{m}_i$. Its powers stabilize, say $R^k = R^{k+1} = N$, with $k \geq 1$, so $N^2 = N$. If $N \neq 0$, choose a minimal ideal J with $NJ \neq 0$, then $x \in J$ with $Nx \neq 0$. Minimality gives $J = (x) = N(x)$, so $x = ax$ for some nilpotent $a \in N$. Iteration forces $x = 0$, a contradiction. Thus a power of R is zero, and the lemma proves Noetherianity. $\square$

48 Structure of Artinian rings

Proposition. *In a Noetherian local ring $(A, \mathfrak{m})$, either all inclusions $\mathfrak{m}^n \supsetneq \mathfrak{m}^{n+1}$ are strict, or $\mathfrak{m}^n = 0$ for some n and A is Artinian.*

Proof. An equality $\mathfrak{m}^n = \mathfrak{m}\mathfrak{m}^n$ implies $\mathfrak{m}^n = 0$ by Nakayama. Every prime then contains $\mathfrak{m}$, so the ring has dimension zero. $\square$

Theorem. *A ring is Artinian if and only if it is a finite product of Artinian local rings. The nonzero local factors are unique up to isomorphism and permutation.*

Proof. For distinct maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ choose k large enough that

$$\bigcap_i \mathfrak{m}_i^k = \prod_i \mathfrak{m}_i^k = 0.$$

The Chinese remainder theorem gives

$$A \simeq \prod_{i=1}^r A/\mathfrak{m}_i^k,$$

whose factors are Artinian local. Conversely a finite product of Artinian rings is Artinian. The kernels of projections to local factors are primary with distinct maximal radicals. They give a minimal primary decomposition of zero, all of whose components correspond to minimal primes. Their uniqueness proves uniqueness of the factors. $\square$

49 The principal ideal theorem

Definition. The height of a prime is $\text{ht } \mathfrak{p} = \dim A_{\mathfrak{p}}$, and the height of a proper ideal is

$$\text{ht } I = \inf_{\mathfrak{p} \supseteq I} \text{ht } \mathfrak{p}.$$

Thus $\dim A = \sup_{\mathfrak{p} \in \text{Spec } A} \text{ht } \mathfrak{p}$.

Theorem (Krull's principal ideal theorem). *If A is Noetherian and $\mathfrak{p}$ is minimal over (x) , then $\text{ht } \mathfrak{p} \leq 1$.*

Proof. Localize at $\mathfrak{p}$; then $A/(x)$ is Artinian local. For $\mathfrak{q} \subsetneq \mathfrak{p}$, it suffices to show $\dim A_{\mathfrak{q}} = 0$. Define the symbolic powers

$$\mathfrak{q}^{(n)} = A \cap \mathfrak{q}^n A_{\mathfrak{q}}.$$

They are $\mathfrak{q}$ -primary: the corresponding quotient embeds into $A_{\mathfrak{q}}/\mathfrak{q}^n A_{\mathfrak{q}}$, where the maximal ideal is nilpotent and every element outside it is a unit. The descending chain $(x) + \mathfrak{q}^{(n)}$ stabilizes modulo (x) . Thus for large n , if $f \in \mathfrak{q}^{(n)}$, write $f = ax + g$ with $g \in \mathfrak{q}^{(n+1)}$. Since $x \notin \mathfrak{q}$, primaryness forces $a \in \mathfrak{q}^{(n)}$. Hence

$$\mathfrak{q}^{(n)} = x\mathfrak{q}^{(n)} + \mathfrak{q}^{(n+1)}.$$

Nakayama applied to the finite quotient gives $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$. Localizing yields $\mathfrak{q}^n A_{\mathfrak{q}} = \mathfrak{q}^{n+1} A_{\mathfrak{q}}$, so $A_{\mathfrak{q}}$ is Artinian local and has dimension zero. $\square$

50 Prime avoidance

Lemma. *Let $I_1, \dots, I_n, J$ be ideals with $J \not\subseteq I_i$ for every i . Suppose either A contains an infinite field, or all but at most two of the I_i are prime. Then*

$$J \not\subseteq I_1 \cup \dots \cup I_n.$$

Proof. In the first case, a vector space over an infinite field is not a finite union of proper subspaces; apply this to J and $J \cap I_i$. For the second, induct on n . If J is already not contained in the union with one ideal removed, choose $x_i \in J \setminus \bigcup_{j \neq i} I_j$. If any $x_i \notin I_i$, it is the desired element; otherwise assume all $x_i \in I_i$. For $n = 2$, use $x_1 + x_2$. For $n \geq 3$, relabel so that I_1 is prime and use

$$x_1 + x_2 x_3 \cdots x_n.$$

Modulo I_1 , the first term is zero and the product is nonzero by primality. Modulo any I_i for $i > 1$, the product is zero while x_1 is nonzero. $\square$

51 The height theorem and its converse

Theorem. *In a Noetherian ring, if $I = (x_1, \dots, x_n)$ and $\mathfrak{p}$ is minimal over I , then $\text{ht } \mathfrak{p} \leq n$. For a proper I , in particular $\text{ht } I \leq n$.*

Proof. Induct on n and localize at $\mathfrak{p}$. For a prime $\mathfrak{p}_1$ maximal among primes properly below $\mathfrak{p}$, some generator, say x_1 , avoids $\mathfrak{p}_1$. Such a maximal intermediate prime exists by the ascending chain condition. The only prime containing $(x_1) + \mathfrak{p}_1$ is $\mathfrak{p}$, so choose a common k and write

$$x_i^k = a_i x_1 + y_i, \quad y_i \in \mathfrak{p}_1 \quad (2 \leq i \leq n).$$

Let $J = (y_2, \dots, y_n)$. Then $\sqrt{(x_1) + J} = \mathfrak{p}$. In A/J , the principal ideal theorem gives height at most one to $\mathfrak{p}/J$, so the smaller prime $\mathfrak{p}_1/J$ has height zero. Thus $\mathfrak{p}_1$ is minimal over J and has height at most $n - 1$ by induction. Every chain below $\mathfrak{p}$ can be extended through such a $\mathfrak{p}_1$, proving the bound. $\square$

Corollary. *Every descending prime chain starting at $\mathfrak{p}$ has finite length, bounded by the number of generators of $\mathfrak{p}$. In $k[x_1, \dots, x_r]$, the ideal $(x_1, \dots, x_r)$ has height r .*

Corollary (Converse). *If $\text{ht } \mathfrak{p} = n$, there exist $x_1, \dots, x_n \in \mathfrak{p}$ such that $\mathfrak{p}$ is minimal over their ideal and that ideal has height n .*

Proof. Inductively choose $x_{r+1} \in \mathfrak{p}$ avoiding the finitely many minimal primes of $(x_1, \dots, x_r)$. These have height r and cannot contain $\mathfrak{p}$ when $r < n$, so prime avoidance applies. Every prime over the enlarged ideal strictly contains one of those minimal primes, giving height at least $r + 1$; the height theorem supplies the reverse bound for minimal primes. At $r = n$, a minimal prime contained in $\mathfrak{p}$ must equal $\mathfrak{p}$. $\square$

52 Systems of parameters and the dimension formula

For a Noetherian local ring $(A, \mathfrak{m})$, its dimension d is finite and equals the smallest number of elements $x_1, \dots, x_d \in \mathfrak{m}$ with

$$\sqrt{(x_1, \dots, x_d)} = \mathfrak{m}.$$

Equivalently their ideal contains a power of $\mathfrak{m}$, or the quotient is Artinian. Such a d -tuple is a *system of parameters*. These assertions follow from the height theorem and its converse.

Theorem. *For a local homomorphism $(A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ of Noetherian local rings,*

$$\dim B \leq \dim A + \dim(B/\mathfrak{m}B).$$

Equality holds if B is flat over A . The closed fiber is displayed in the Cartesian square

$$\begin{array}{ccc} \operatorname{Spec}(B/\mathfrak{m}B) & \longrightarrow & \operatorname{Spec} B \\ \downarrow & & \downarrow \\ \operatorname{Spec}(A/\mathfrak{m}) & \longrightarrow & \operatorname{Spec} A. \end{array}$$

Proof. Choose parameters $x_1, \dots, x_d$ in A and lifts $y_1, \dots, y_e$ of parameters in $B/\mathfrak{m}B$. Powers of $\mathfrak{m}$ and $\mathfrak{n}$ show that $(x_1, \dots, x_d, y_1, \dots, y_e)B$ is $\mathfrak{n}$ -primary. The height theorem gives $\dim B \leq d + e$. If B is flat, choose a minimal prime $\mathfrak{q}$ over $\mathfrak{m}B$ with $\dim(B/\mathfrak{q}) = e$. Going down lifts a chain of length d below $\mathfrak{m}$ to a chain below $\mathfrak{q}$. Concatenating with a chain of length e above $\mathfrak{q}$ gives $\dim B \geq d + e$. $\square$

Lemma (Going down for flat maps). *If $A \rightarrow B$ is flat, $\mathfrak{p}' \subseteq \mathfrak{p}$, and $\mathfrak{q}$ contracts to $\mathfrak{p}$, then some $\mathfrak{q}' \subseteq \mathfrak{q}$ contracts to $\mathfrak{p}'$.*

Proof in the Noetherian setting used above. Pass to $A/\mathfrak{p}' \rightarrow B/\mathfrak{p}'B$, with base now a domain, and choose a minimal prime $\mathfrak{q}' \subseteq \mathfrak{q}$. A minimal prime is associated, so each of its elements is a zerodivisor. Flatness makes the target torsion-free over the domain, hence $\mathfrak{q}' \cap A = 0$. $\square$

53 Completion and polynomial dimension

Corollary. *For a Noetherian local ring, $\dim \widehat{A} = \dim A$.*

Proof. The completion is Noetherian local and flat, with maximal ideal $\mathfrak{m}\widehat{A}$ and residue field $A/\mathfrak{m}$. The dimension formula has zero-dimensional closed fiber. $\square$

Corollary. *If A is a nonzero Noetherian ring, then*

$$\dim A[x] = \dim A + 1.$$

For a field k , $\dim k[x_1, \dots, x_n] = n$.

Proof. For $\mathfrak{p} \in \operatorname{Spec} A$, primes of $A[x]$ over $\mathfrak{p}$ correspond to primes of $\kappa(\mathfrak{p})[x]$. A prime $\mathfrak{q}$ maximal in this fiber has one-dimensional local fiber, so flatness and the local dimension formula give

$$\dim A[x]_{\mathfrak{q}} = \dim A_{\mathfrak{p}} + 1.$$

For an arbitrary prime $\mathfrak{q}$, the fiber dimension is at most one, giving the upper bound after taking suprema. Such maximal fiber primes exist for every $\mathfrak{p}$, giving the lower bound. Induct on the number of variables for the field case. $\square$

Regular sequences

A sequence $x_1, \dots, x_r$ is M -regular if each x_i acts injectively on the preceding quotient

$$M/(x_1, \dots, x_{i-1})M.$$

When speaking of regular sequences in a local ring below, the elements lie in the maximal ideal and M is nonzero finite, so the final quotient is nonzero by Nakayama.

54 Regular local rings

Definition. Let $(A, \mathfrak{m}, k)$ be Noetherian local of dimension d . The height theorem shows that $\mathfrak{m}$ needs at least d generators. The ring is *regular local* if $\mathfrak{m}$ can be generated by d elements, equivalently

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = d.$$

The equivalence follows from Nakayama. A minimal generating d -tuple is called a regular system of parameters.

Proposition. *A regular local ring is a domain.*

Proof. Induct on d . For $d = 0$, $\mathfrak{m} = 0$ and A is a field. Otherwise choose $x \in \mathfrak{m}$ outside $\mathfrak{m}^2$ and every minimal prime, using prime avoidance. This is possible because $\mathfrak{m} \neq \mathfrak{m}^2$ and $\mathfrak{m}$ is not minimal when $d > 0$. Put $B = A/(x)$ with maximal ideal $\mathfrak{n}$. Avoidance of minimal primes gives $\dim B < d$, while lifting parameters of B and adjoining x gives $d \leq \dim B + 1$. Thus $\dim B = d - 1$. Moreover

$$\dim_k \mathfrak{n}/\mathfrak{n}^2 = d - 1,$$

since $x \notin \mathfrak{m}^2$. Therefore B is regular and by induction a domain, so (x) is prime. Choose a minimal prime $\mathfrak{q} \subseteq (x)$. Since $x \notin \mathfrak{q}$, any $u = ax \in \mathfrak{q}$ has $a \in \mathfrak{q}$. Hence $\mathfrak{q} = x\mathfrak{q}$, and Nakayama gives $\mathfrak{q} = 0$. This makes zero prime, proving the assertion. $\square$

Remark. The fact that a regular system of parameters is a regular sequence requires an argument in addition to the definition. One may apply the domain result and induction to successive parameter quotients.

55 Discrete valuation rings

Definition. A discrete valuation on a field K is a homomorphism $v : K^\times \rightarrow \mathbb{Z}$ satisfying

$$v(a + b) \geq \min\{v(a), v(b)\}, \quad v(0) = \infty.$$

Its valuation ring is $\{a : v(a) \geq 0\}$, including zero. A *discrete valuation ring* (DVR) is the valuation ring of a nontrivial such valuation; rescaling by the positive generator of its image normalizes the image to $\mathbb{Z}$.

Example. For a prime p , $v_p(p^n a/b) = n$ when a, b are prime to p . Its ring is $\mathbb{Z}_{(p)} = \{a/b : p \nmid b\}$.

Theorem. *A ring is a DVR if and only if it is a regular local ring of dimension one.*

Proof. For a one-dimensional regular local ring, write $\mathfrak{m} = (\pi)$. It is a domain. Krull intersection shows that every nonzero $a \in A$ has a largest order n , so $a = u\pi^n$ with u a unit. Every $f \in \text{Frac}(A)^\times$ therefore has a unique expression $u\pi^n$, $n \in \mathbb{Z}$. Defining $v(f) = n$ gives the valuation inequality by factoring out the smaller power from a sum. Every nonzero ideal is generated by an element of least order, so has the form (π^n) . Conversely, in a valuation ring the positive-value elements form the unique maximal ideal. Choose π of least positive value. For any nonzero ideal, choose an element of least value; division shows it generates that ideal. All values are multiples of $v(\pi)$, so all nonzero ideals are (π^n) . The ring is a local PID with exactly the primes (0) and (π) , hence regular of dimension one. $\square$

56 Depth and Serre's conditions

Definition. The *depth* of a Noetherian local ring $(A, \mathfrak{m})$ is the maximum length of an A -regular sequence in $\mathfrak{m}$.

Proposition. For $f \in \mathfrak{m}$,

$$\dim A - 1 \leq \dim A/(f) \leq \dim A.$$

If f is A -regular, then $\dim A/(f) = \dim A - 1$. Consequently $\text{depth } A \leq \dim A$.

Proof. Lift parameters of $A/(f)$ and adjoin f to obtain the lower bound; quotienting gives the upper bound. A regular element avoids all minimal primes, so each prime chain in $A/(f)$ can be lengthened below its first term in A . Thus the dimension drops strictly, hence by exactly one. Iterate along a regular sequence. $\square$

For a Noetherian ring and $r \geq 0$, Serre's conditions are

$$\begin{aligned} (R_r) : & \quad A_{\mathfrak{p}} \text{ is regular whenever } \text{ht } \mathfrak{p} \leq r, \\ (S_r) : & \quad \text{depth } A_{\mathfrak{p}} \geq \min\{r, \text{ht } \mathfrak{p}\} \text{ for every } \mathfrak{p}. \end{aligned}$$

A ring is *regular* if every prime localization is regular local; it is *Cohen–Macaulay* if $\text{depth } A_{\mathfrak{p}} = \dim A_{\mathfrak{p}}$ for all $\mathfrak{p}$. A ring is *normal* if all prime localizations are normal domains.

Theorem. Regularity is equivalent to all (R_r) ; Cohen–Macaulayness to all (S_r) . Reducedness is equivalent to (R_0) and (S_1) (exercise). Normality is equivalent to (R_1) and (S_2) , Serre's criterion proved below.

57 The total quotient ring

Proposition. *Let A be Noetherian and reduced, and $z \in Q(A)$. Then $z \in A$ if and only if, for every nonzerodivisor $s \in A$ and every $\mathfrak{p} \in \text{Ass}(A/(s))$, the image of z in $Q(A)_{\mathfrak{p}}$ belongs to the image of $A_{\mathfrak{p}}$.*

Proof. If $z = a/s \notin A$, the class of a in $A/(s)$ is nonzero. The injection into the product of its associated-prime localizations gives a $\mathfrak{p}$ where that class survives. Thus $a/1 \notin sA_{\mathfrak{p}}$, and a/s does not belong to $A_{\mathfrak{p}}$. $\square$

The localization maps involved are

$$\begin{array}{ccc} A & \longrightarrow & A_{\mathfrak{p}} \\ \downarrow & & \downarrow \\ Q(A) & \longrightarrow & Q(A)_{\mathfrak{p}}. \end{array}$$

For a domain, this becomes an intersection inside its fraction field:

$$A = \bigcap_{\mathfrak{p} \in \mathcal{C}} A_{\mathfrak{p}}, \quad \mathcal{C} = \bigcup_{0 \neq s \in A} \text{Ass}(A/(s)).$$

A DVR is integrally closed. Indeed, if z has negative valuation, in a monic relation $z^n + a_1 z^{n-1} + \cdots + a_n = 0$ the term z^n has strictly smaller valuation than all others, which is impossible.

58 Normal domains and codimension one

Theorem. *For a Noetherian domain A , the following are equivalent:*

1. A is normal.
2. For every $0 \neq s \in A$ and $\mathfrak{p} \in \text{Ass}(A/(s))$, the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ is principal.
3. All these $A_{\mathfrak{p}}$ are DVRs.
4. A satisfies (R_1) and (S_2) .

Proof. For $(1) \Rightarrow (2)$, localize at such a $\mathfrak{p}$, and choose $b \notin (s)$ with $\mathfrak{p} = \text{Ann}(b \bmod s)$. Put

$$\mathfrak{p}^{-1} = \{z \in \text{Frac } A : z\mathfrak{p} \subseteq A\}.$$

Then $\mathfrak{p} \subseteq \mathfrak{p}^{-1}\mathfrak{p} \subseteq A$. If $\mathfrak{p}^{-1}\mathfrak{p} \subseteq \mathfrak{p}$, multiplication by each $z \in \mathfrak{p}^{-1}$ preserves the finite faithful module $\mathfrak{p}$. The determinant trick makes z integral, hence in A . But $b/s \in \mathfrak{p}^{-1} \setminus A$, a contradiction. Thus $\mathfrak{p}^{-1}\mathfrak{p} = A$. In a local ring some product zu in an expression $1 = \sum z_i u_i$ is a unit. For $v \in \mathfrak{p}$, the identity $v = u(zv)/(zu)$ shows $\mathfrak{p} = (u)$. The principal ideal theorem and the DVR characterization give $(2) \Leftrightarrow (3)$. For $(3) \Rightarrow (1)$, use the preceding intersection description and normality of DVRs. For $(3) \Rightarrow (4)$, a height-one localization is detected by a nonzero element of its maximal ideal. At a localization of height at least two, choose a nonzero s in its maximal ideal. Its associated primes modulo s have height one by (3); prime avoidance gives a second regular element. Conversely, under (S_2) a prime associated to $A/(s)$ has depth one, hence height one; (R_1) makes its localization a DVR. $\square$

59 Serre's criterion for normality

Proposition. *For a Noetherian ring, the following are equivalent: normality; being a finite product of normal domains; being reduced and integrally closed in its total quotient ring.*

Theorem (Serre). *A Noetherian ring is normal if and only if it satisfies (R_1) and (S_2) .*

Proof. For a finite product of normal domains, the result follows from the domain case: its spectrum is the disjoint union of the spectra of its factors, and localizations are those of the appropriate factor. Conversely assume (R_1) and (S_2) . In a minimal primary decomposition $0 = \cap_i Q_i$, each $\mathfrak{p}_i = \sqrt{Q_i}$ is associated. The local ring $A_{\mathfrak{p}_i}$ has depth zero, so (S_1) forces $\text{ht } \mathfrak{p}_i = 0$. By (R_0) it is a field. Uniqueness of isolated components gives $Q_i = \ker(A \rightarrow A_{\mathfrak{p}_i}) = \mathfrak{p}_i$. Thus A is reduced. Let $z \in Q(A)$ be integral over A . For a nonzerodivisor s and $\mathfrak{p} \in \text{Ass}(A/(s))$, the depth drop gives $\text{depth } A_{\mathfrak{p}} = 1$. Condition (S_2) forces $\text{ht } \mathfrak{p} = 1$, and (R_1) makes $A_{\mathfrak{p}}$ a DVR. Hence the image of z lies in $A_{\mathfrak{p}}$. The preceding detection proposition gives $z \in A$, proving integral closedness in $Q(A)$ and therefore normality. $\square$

60 Going down over a normal domain

Lemma. *If A is a normal domain, a monic polynomial irreducible in $A[t]$ is irreducible in $\text{Frac}(A)[t]$, and generates a prime ideal of $A[t]$.*

Proof. In a factorization into monic factors over the fraction field, coefficients of the factors are symmetric expressions in roots integral over A . They are integral and lie in $\text{Frac } A$, hence lie in A . This proves irreducibility over the fraction field. The quotient $A[t]/(f)$ is free over A and embeds in the field $\text{Frac}(A)[t]/(f)$, hence is a domain. $\square$

Theorem. *An integral extension of domains $A \subseteq B$ satisfies going down if A is normal.*

Proof. Let $\mathfrak{p}' \subseteq \mathfrak{p}$ and $\mathfrak{q} \cap A = \mathfrak{p}$. It suffices to prove $A \cap \mathfrak{p}'B_{\mathfrak{q}} = \mathfrak{p}'$: then the fiber $B_{\mathfrak{q}} \otimes_A \kappa(\mathfrak{p}')$ is nonzero and supplies the required prime. Take $a \in A \cap \mathfrak{p}'B_{\mathfrak{q}}$, say $a = b/s$ with $b \in \mathfrak{p}'B$, $s \notin \mathfrak{q}$. If $a = 0$ there is nothing to prove. The element b satisfies a monic equation $f(t)$ with its degree- j coefficient in $(\mathfrak{p}')^j$: apply the determinant trick in a finite subalgebra containing the finitely many elements needed to express b . Let $g(t) = t^n + a_1t^{n-1} + \cdots + a_n$ be the monic minimal polynomial of s . Its coefficients lie in A by normality. Not all a_j lie in $\mathfrak{p}'$, since otherwise $s^n \in \mathfrak{p}'B \subseteq \mathfrak{q}$. Since $b = as$, its minimal polynomial is

$$r(t) = t^n + aa_1t^{n-1} + \cdots + a^na_n.$$

It divides f in $A[t]$, by monic division. Modulo $\mathfrak{p}'$, it divides a power of t ; passing to the fraction field of $A/\mathfrak{p}'$ shows its lower coefficients vanish. Thus $a^ja_j \in \mathfrak{p}'$ for every j . Since some $a_j \notin \mathfrak{p}'$, primality gives $a \in \mathfrak{p}'$. $\square$

61 Coordinate changes for normalization

Definition. An *affine k -algebra* is a finitely generated k -algebra.

Lemma. For a nonconstant $f \in k[x_1, \dots, x_n]$, there are $y_1, \dots, y_{n-1}$ such that $k[x_1, \dots, x_n]$ is finite over $k[y_1, \dots, y_{n-1}, f]$. One may choose $y_i = x_i - x_n^{e_i}$ for sufficiently large e_i . If k is infinite, one may instead choose $y_i = x_i - a_i x_n$ for suitable $a_i \in k$.

Proof. In the first change of variables, a monomial $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ has leading x_n -degree

$$\alpha_1 e + \alpha_2 e^2 + \cdots + \alpha_{n-1} e^{n-1} + \alpha_n.$$

For e larger than the exponents occurring in f , these degrees are distinct. Hence f becomes a polynomial in x_n with nonzero constant leading coefficient. Divide by that scalar; the equation expressing f makes x_n integral over $k[y_1, \dots, y_{n-1}, f]$. For the second change, if f_d is the top homogeneous part, the leading coefficient in x_n is $f_d(a_1, \dots, a_{n-1}, 1)$. Choose the a_i to make it nonzero, possible over an infinite field. The same argument applies. $\square$

Theorem (Noether normalization). *Every nonzero affine k -algebra A is finite over a polynomial subalgebra $k[z_1, \dots, z_d]$, where $d = \dim A$.*

Proof. Induct on the number of algebra generators. If they are independent, there is nothing to prove. Otherwise apply the coordinate-change lemma to a nonzero polynomial relation: its image makes the last generator integral over an algebra on one fewer generator. Iterate and use transitivity of finiteness. Dimension invariance under integral extension identifies the number of independent remaining generators with $\dim A$. $\square$

62 Normalization adapted to a chain of ideals

Theorem. *Let A be a nonzero affine k -algebra of dimension d , and $I_1 \subseteq \cdots \subseteq I_m$ proper ideals. Put $d_j = \dim(A/I_j)$. There is a polynomial subalgebra $B = k[x_1, \dots, x_d] \subseteq A$, finite under A , such that*

$$I_j \cap B = (x_{d_j+1}, \dots, x_d) \quad (1 \leq j \leq m).$$

Geometrically, writing $V = \operatorname{Spec} A$ and $V_j = \operatorname{Spec}(A/I_j)$ gives finite surjections in the commuting diagram

$$\begin{array}{ccccccc} V & \longleftarrow & V_1 & \longleftarrow & \cdots & \longleftarrow & V_m \\ \downarrow & & \downarrow & & & & \downarrow \\ \mathbb{A}_k^d & \longleftarrow & \mathbb{A}_k^{d_1} & \longleftarrow & \cdots & \longleftarrow & \mathbb{A}_k^{d_m}. \end{array}$$

Proof. First use ordinary normalization to make A finite over $k[y_1, \dots, y_d]$. Modify this polynomial subalgebra recursively. At stage e , retain coordinates $x_{e+1}, \dots, x_d$ and choose $x'_1, \dots, x'_e$ so that A is finite over their polynomial algebra and

$$I_j \supseteq (x_{\max(e, d_j)+1}, \dots, x_d).$$

If $e = d_m$, stop. Otherwise choose the first j with $d_j < e$. Then $I_j \cap k[x'_1, \dots, x'_e]$ contains a nonzero nonconstant polynomial, since the finite image has dimension $d_j < e$ and the tail coordinates already vanish modulo I_j . Take such an element as x_e . The coordinate-change lemma replaces the other $e - 1$ coordinates while maintaining finiteness. Since $I_j \subseteq I_l$ for $l \geq j$, the containment condition persists for every ideal. Continue with $e - 1$. At the end $J_j = (x_{d_j+1}, \dots, x_d) \subseteq I_j \cap B$. The integral extension of quotients gives $\dim B/(I_j \cap B) = d_j$. The quotient B/J_j is a polynomial ring of dimension d_j ; a nonzero ideal in this domain lowers dimension. Hence $I_j \cap B = J_j$. $\square$

63 Dimensions of affine rings

Definition. For a domain A over k , the transcendence degree $\text{trdeg}_k A$ is the cardinality of a transcendence basis of $\text{Frac } A/k$, equivalently the supremum of sizes of algebraically independent subsets of A . For example, $\text{trdeg}_k k[x_1, \dots, x_n] = n$.

Theorem. For an affine domain A/k ,

$$\dim A = \text{trdeg}_k A.$$

Proof. Choose a finite normalization $k[x_1, \dots, x_d] \subseteq A$. Integral extension preserves dimension, and the fraction-field extension is algebraic, so both sides equal d . $\square$

Corollary. For an inclusion of affine domains $A \subseteq B$ over k ,

$$\dim B = \dim A + \dim(\text{Frac}(A) \otimes_A B).$$

Proof. The generic fiber is an affine domain over $\text{Frac } A$. Apply the theorem and additivity of transcendence degree in a tower of fields. $\square$

Theorem. Every maximal chain of primes in an affine domain A has length $\dim A$.

Proof. Let $0 = \mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_r$ be a maximal chain. Choose normalization adapted to these ideals, so $\mathfrak{p}_i \cap B = (x_{d_i+1}, \dots, x_d)$, where $d_i = \dim(A/\mathfrak{p}_i)$ and $d_0 = d$. Writing $J_i = (x_{d_i+1}, \dots, x_d)$, the chain and its contractions are

$$\begin{array}{ccccccc} \mathfrak{p}_0 & \hookrightarrow & \mathfrak{p}_1 & \hookrightarrow & \dots & \hookrightarrow & \mathfrak{p}_r \\ \downarrow & & \downarrow & & & & \downarrow \\ J_0 & \hookrightarrow & J_1 & \hookrightarrow & \dots & \hookrightarrow & J_r. \end{array}$$

Incomparability makes the d_i strictly decrease; the final maximal ideal has $d_r = 0$. If $d_i - d_{i+1} > 1$, insert an intermediate coordinate prime between the contractions. The quotient $B/(\mathfrak{p}_i \cap B)$ is a polynomial ring, hence normal. Going down in the integral domain extension to $A/\mathfrak{p}_i$ lifts the intermediate prime below $\mathfrak{p}_{i+1}/\mathfrak{p}_i$, contradicting maximality. Thus every drop is one, so $r = d$. $\square$

64 Generic freeness

Theorem (Generic freeness). *Let A be a Noetherian domain, let B be a finitely generated A -algebra, and let M be a finitely generated B -module. There is a nonzero $f \in A$ such that M_f is free as an A_f -module.*

Proof sketch. Induct on $\dim B$. If a nonzero element of A annihilates B , localization finishes the proof. Otherwise $A \hookrightarrow B$. Noether normalization over $K = \text{Frac } A$ gives algebraically independent elements $x_1, \dots, x_d$ for which $K \otimes_A B$ is finite over $K[x_1, \dots, x_d]$. After clearing denominators, localize A so that B is finite over $A[x_1, \dots, x_d]$. Choose a finite filtration of M whose factors are cyclic. For a factor $B/\mathfrak{q}$, either $\mathfrak{q} \cap A \neq 0$, in which case a further localization kills it, or $B/\mathfrak{q}$ is generically finite over a polynomial ring and becomes free after clearing finitely many coefficients. The extensions split over the polynomial ring after one more localization, so the entire filtered module becomes free over A_f . $\square$

Corollary. *If $A \subseteq B$ is an inclusion of finitely generated domains, then for some $0 \neq f \in A$ the morphism $\text{Spec } B_f \rightarrow \text{Spec } A_f$ is surjective. For $\mathfrak{q} \in \text{Spec } B_f$ and $\mathfrak{p} = \mathfrak{q} \cap A_f$,*

$$\dim B_{\mathfrak{q}} = \dim A_{\mathfrak{p}} + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

65 Dimension of fibers

Let k be a field and $A \subseteq B$ an inclusion of affine k -domains. Put

$$d = \operatorname{trdeg}_{\operatorname{Frac} A} \operatorname{Frac} B.$$

Generic freeness supplies $0 \neq f \in A$ such that every maximal ideal $\mathfrak{m}$ of A_f has a prime of B_f above it and every irreducible component of the fiber has dimension d .

Theorem (Upper semicontinuity of fiber dimension). *For a dominant morphism $X \rightarrow Y$ of irreducible affine varieties, there is a nonempty open set $U \subseteq Y$ such that every irreducible component of every fiber over U has dimension $\dim X - \dim Y$.*

Equivalently, for a finite-type map $A \rightarrow B$ of Noetherian affine rings and an integer r , the locus where the fiber dimension is at least r is closed after restricting to each irreducible component of the base. The local inequality behind this statement is

$$\dim B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}} \geq \dim B_{\mathfrak{q}} - \dim A_{\mathfrak{p}}, \quad \mathfrak{p} = \mathfrak{q} \cap A.$$

The proof reduces first to domains and then localizes at the generic point. Generic freeness handles a dense open set; Noetherian induction treats its closed complement. This also explains why the theorem describes a generic open set rather than every fiber.

66 Ext and its exact sequences

Definition. Let N, M be A -modules and $P_\bullet \rightarrow N$ a projective resolution. Define

$$\mathrm{Ext}_A^i(N, M) = H^i(\mathrm{Hom}_A(P_\bullet, M)).$$

This is independent of the chosen resolution. In particular,

$$\mathrm{Ext}_A^0(N, M) = \mathrm{Hom}_A(N, M), \quad \mathrm{Ext}_A^i(P, M) = 0 \quad (i > 0)$$

when P is projective. If A is Noetherian and M, N are finite, then every $\mathrm{Ext}_A^i(N, M)$ is finite.

A short exact sequence in either variable produces a long exact sequence. For example, from $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ one obtains

$$0 \rightarrow \mathrm{Hom}(N, M') \rightarrow \mathrm{Hom}(N, M) \rightarrow \mathrm{Hom}(N, M'') \rightarrow \mathrm{Ext}^1(N, M') \rightarrow \cdots .$$

If I is an ideal, then

$$\mathrm{Ext}_A^i(A/I, M) \cong H^i(\mathrm{Hom}_A(P_\bullet, M))$$

for any projective resolution of A/I . These groups will detect regular sequences and depth.

67 Regular sequences and Ext

Definition. Let M be a finite module over a Noetherian local ring $(A, \mathfrak{m})$. A sequence $x_1, \dots, x_n \in \mathfrak{m}$ is M -regular if multiplication by x_i is injective on $M/(x_1, \dots, x_{i-1})M$ for every i and the final quotient is nonzero.

Proposition. If $x_1, \dots, x_n$ is M -regular and $I = (x_1, \dots, x_n)$, then

$$\mathrm{Ext}_A^i(A/I, M) = 0 \quad (i < n), \quad \mathrm{Ext}_A^n(A/I, M) \cong M/IM.$$

More generally, for any A/I -module N ,

$$\mathrm{Ext}_A^{i+n}(N, M) \cong \mathrm{Ext}_{A/I}^i(N, M/IM).$$

The case $n = 1$ follows from $0 \rightarrow A \xrightarrow{x_1} A \rightarrow A/(x_1) \rightarrow 0$; induction gives the general case.

Define the depth of M along an ideal I by the common length of its maximal M -regular sequences contained in I . Then

$$\mathrm{depth}_I M = \inf\{i : \mathrm{Ext}_A^i(A/I, M) \neq 0\}.$$

For a local ring, $\mathrm{depth} M = \mathrm{depth}_{\mathfrak{m}} M$. Localization gives

$$\mathrm{depth}_I M = \inf_{\mathfrak{p} \supseteq I} \mathrm{depth}_{A_{\mathfrak{p}}} M_{\mathfrak{p}}.$$

This Ext formulation proves that all maximal regular sequences in I have the same length.

68 Depth, support, and dimension

For finite modules over a Noetherian local ring,

1. $\text{depth } M = 0$ exactly when $\mathfrak{m} \in \text{Ass}(M)$;
2. $\text{depth}(M \oplus N) = \min\{\text{depth } M, \text{depth } N\}$;
3. in $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, each depth is bounded below by the minimum of the other two, with the usual strict-case refinements (the depth lemma);
4. $\text{depth } M \leq \dim M := \dim \text{Supp } M$.

If $x \in \mathfrak{m}$ is M -regular, then

$$\text{depth}(M/xM) = \text{depth } M - 1, \quad \dim(M/xM) = \dim M - 1$$

provided x is chosen outside the minimal primes of $\text{Supp } M$. If M is finite and $\text{Supp } M = V(I)$, then

$$\text{depth}_I M = \inf_{\mathfrak{p} \in V(I)} \text{depth } M_{\mathfrak{p}}.$$

The dimension of M is also the least number of parameters needed to make $M/(x_1, \dots, x_r)M$ have finite length. Prime avoidance lets one choose parameters successively outside the minimal primes of the current support.

69 Projective dimension and minimal resolutions

Definition. The projective dimension of an A -module M is

$$\mathrm{pd}_A M = \inf\{n : M \text{ has a projective resolution of length } n\}.$$

The global dimension is $\mathrm{gldim} A = \sup_M \mathrm{pd}_A M$.

If $(A, \mathfrak{m}, k)$ is local and M is finite, a free resolution $F_\bullet \rightarrow M$ is minimal when every differential has image in $\mathfrak{m}F_{i-1}$. Minimal resolutions exist and are unique up to isomorphism. Tensoring one with k kills every differential, hence

$$\mathrm{Tor}_i^A(k, M) \cong k \otimes_A F_i.$$

Consequently

$$\mathrm{pd}_A M = \sup\{i : \mathrm{Tor}_i^A(k, M) \neq 0\} = \sup\{i : \mathrm{Ext}_A^i(M, k) \neq 0\}.$$

If $M \neq 0$ has finite projective dimension, the last nonzero free module of a minimal resolution cannot disappear after localization at $\mathfrak{m}$.

70 The Auslander–Buchsbaum formula

Theorem (Auslander–Buchsbaum). *Let $(A, \mathfrak{m}, k)$ be Noetherian local and let $0 \neq M$ be finite with $\mathrm{pd}_A M < \infty$. Then*

$$\mathrm{pd}_A M + \mathrm{depth}_A M = \mathrm{depth} A.$$

Proof sketch. Induct on $\mathrm{depth} A$. If $\mathrm{depth} A = 0$, then $\mathfrak{m} \in \mathrm{Ass} A$. A minimal resolution ending in $F_n \neq 0$ cannot have $n > 0$: an element annihilated by $\mathfrak{m}$ produces a nonzero kernel vector in F_n , contradicting exactness. Thus M is free.

If $\mathrm{depth} A > 0$ and $\mathrm{depth} M > 0$, choose $x \in \mathfrak{m}$ regular on both A and M by prime avoidance. Reduction modulo x preserves projective dimension and lowers both depths by one; induction applies over $A/(x)$.

If $\mathrm{depth} M = 0$, use the first syzygy $0 \rightarrow M_1 \rightarrow F \rightarrow M \rightarrow 0$. The depth lemma gives $\mathrm{depth} M_1 = 1$, and $\mathrm{pd} M_1 = \mathrm{pd} M - 1$. Apply the preceding case to M_1 . $\square$

Corollary. *For a regular local ring $(A, \mathfrak{m}, k)$, $\mathrm{pd}_A k = \dim A$. In particular its global dimension is $\dim A$.*

71 Koszul complexes

Definition. Let M be an A -module. The tensor algebra $T(M) = \bigoplus_{n \geq 0} M^{\otimes n}$ has the exterior algebra

$$\bigwedge M = T(M)/(x \otimes x : x \in M)$$

as a quotient. If M is free on $e_1, \dots, e_r$, then $\bigwedge^n M$ is free on $e_{i_1} \wedge \dots \wedge e_{i_n}$ with $i_1 < \dots < i_n$.

For $x = (x_1, \dots, x_r) \in A^r$, the Koszul complex $K_\bullet(x; A)$ has

$$K_n = \bigwedge^n A^r, \quad d(e_{i_1} \wedge \dots \wedge e_{i_n}) = \sum_j (-1)^{j-1} x_{i_j} e_{i_1} \wedge \dots \wedge \widehat{e}_{i_j} \wedge \dots \wedge e_{i_n}.$$

For an A -module M , put $K_\bullet(x; M) = K_\bullet(x; A) \otimes_A M$. Multiplication by every x_i is null-homotopic, so the homology $H_i(x; M)$ is naturally an $A/(x)$ -module. Moreover

$$H_0(x; M) = M/(x)M.$$

The self-duality of the exterior algebra identifies the cochain Koszul complex with a shift of the chain complex.

72 Koszul homology and regular sequences

The Koszul complex satisfies

$$K(x_1, \dots, x_r; M) \cong K(x_1; A) \otimes \cdots \otimes K(x_r; A) \otimes M.$$

From the mapping-cone description one gets exact sequences relating the homology before and after adjoining one element.

Theorem. *For $x_1, \dots, x_r \in \mathfrak{m}$ and a nonzero finite module M over a local ring, the following are equivalent:*

1. $x_1, \dots, x_r$ is an M -regular sequence;
2. $H_i(x; M) = 0$ for every $i > 0$;
3. $H_1(x; M) = 0$.

Thus a regular sequence makes $K_\bullet(x; A)$ a free resolution of $A/(x)$. If $(A, \mathfrak{m}, k)$ is regular local and x is a regular system of parameters, the Koszul resolution gives

$$\mathrm{pd}_A k \leq \dim A.$$

Combined with Auslander–Buchsbaum, equality holds.

73 Cohen–Macaulay modules

Definition. Let $(A, \mathfrak{m})$ be Noetherian local. A nonzero finite module M is Cohen–Macaulay if

$$\text{depth } M = \dim M.$$

The zero module is conventionally Cohen–Macaulay. A ring is Cohen–Macaulay when it is so as a module over itself.

Proposition. *If M is Cohen–Macaulay of dimension d , then every system of parameters for M is an M -regular sequence. Conversely, if one system of parameters is regular, then M is Cohen–Macaulay.*

If $\mathfrak{p} \in \text{Supp } M$, localization preserves Cohen–Macaulayness and

$$\dim M_{\mathfrak{p}} + \dim A/\mathfrak{p} = \dim M$$

when M is Cohen–Macaulay. In particular, every associated prime of M is minimal in its support: Cohen–Macaulay modules are unmixed.

For $x \in \mathfrak{m}$ regular on M ,

$$M \text{ is Cohen–Macaulay} \iff M/xM \text{ is Cohen–Macaulay.}$$

If $A \rightarrow B$ is finite local and M is a finite B -module, then depth and dimension computed over A agree with those computed over B . Polynomial rings and their localizations at homogeneous maximal ideals are Cohen–Macaulay.

74 Unmixedness

Let $(A, \mathfrak{m})$ be Noetherian local and M Cohen–Macaulay. If

$$0 = \bigcap_i Q_i$$

is a minimal primary decomposition in M , then every $\sqrt{\text{Ann}(M/Q_i)}$ has dimension $\dim M$. Hence there are no embedded associated primes.

One proof chooses a parameter sequence avoiding the finitely many minimal primes. Successive regularity keeps the expected dimension drop. Localizing at an associated prime forces depth zero, and Cohen–Macaulayness then forces local dimension zero. The dimension formula yields

$$\dim A/\mathfrak{p} = \dim M.$$

This result is often called the unmixedness theorem for Cohen–Macaulay modules. Its converse fails without additional hypotheses: absence of embedded primes alone does not force the full depth equalities.

75 Regular local rings and finite global dimension

Theorem (Auslander–Buchsbaum–Serre). *For a Noetherian local ring $(A, \mathfrak{m}, k)$, the following are equivalent:*

1. A is regular;
2. $\mathrm{pd}_A k < \infty$;
3. A has finite global dimension.

When these hold, $\mathrm{gldim} A = \dim A$.

If A is regular and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$, then $A/(x)$ is regular.

If $x_1, \dots, x_d$ is a regular system of parameters, the associated graded ring is

$$\mathrm{gr}_{\mathfrak{m}} A \cong k[X_1, \dots, X_d], \quad X_i \longmapsto x_i \bmod \mathfrak{m}^2.$$

Surjectivity follows because the initial forms generate; equality of dimensions, or comparison of Hilbert functions via the Koszul resolution, gives injectivity. Consequently the $\mathfrak{m}$ -adic order behaves like total degree.

76 Regular rings are UFDs

Theorem. *Every regular local ring is a unique factorization domain.*

Equivalently, every height-one prime of a regular local ring is principal. We use this standard theorem without proof. Since a Noetherian domain is a UFD exactly when every height-one prime is principal, it says geometrically that codimension-one irreducible closed subsets of the spectrum of a regular local ring are locally defined by one equation.

77 Injective modules

Definition. An A -module I is injective if $\mathrm{Hom}_A(-, I)$ is exact. The following are equivalent:

1. I is injective;
2. $\mathrm{Ext}_A^1(M, I) = 0$ for every M ;
3. every map from an ideal $J \subseteq A$ to I extends to a map $A \rightarrow I$ (Baer's criterion).

Every module embeds in an injective module. An injective resolution

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

may therefore be used to compute Ext in the second variable:

$$\mathrm{Ext}_A^i(N, M) = H^i \mathrm{Hom}_A(N, I^\bullet).$$

Direct sums of injectives are injective over a Noetherian ring. Localization of an injective is injective over the localized ring, and for finite N ,

$$\mathrm{Ext}_A^i(N, M)_{\mathfrak{p}} \cong \mathrm{Ext}_{A_{\mathfrak{p}}}^i(N_{\mathfrak{p}}, M_{\mathfrak{p}}).$$

Hence $\mathrm{Supp} \mathrm{Ext}_A^i(N, M) \subseteq \mathrm{Supp} N \cap \mathrm{Supp} M$.

78 Injective dimension and depth

Definition. The injective dimension $\mathrm{id}_A M$ is the least length of an injective resolution. For a finite module over a Noetherian local ring,

$$\mathrm{depth} M = \inf\{i : \mathrm{Ext}_A^i(k, M) \neq 0\}.$$

If M has finite injective dimension, then

$$\mathrm{id}_A M = \sup\{\mathrm{depth} A_{\mathfrak{p}} : \mathfrak{p} \in \mathrm{Supp} M\}.$$

In particular, for a local ring and a nonzero finite M of finite injective dimension,

$$\mathrm{id}_A M = \mathrm{depth} A.$$

The proof localizes a minimal injective resolution. Its terms decompose into injective hulls of residue fields, and $\mathrm{Ext}_{A_{\mathfrak{p}}}^i(\kappa(\mathfrak{p}), M_{\mathfrak{p}})$ records the multiplicity of $E(A/\mathfrak{p})$ in degree i . Nonvanishing first occurs at $\mathrm{depth} M_{\mathfrak{p}}$.

79 Gorenstein local rings

Definition. A Noetherian local ring $(A, \mathfrak{m}, k)$ of dimension d is Gorenstein if

$$\mathrm{id}_A A < \infty.$$

Equivalently, $\mathrm{id}_A A = d$. Every regular local ring is Gorenstein, and every Gorenstein local ring is Cohen–Macaulay.

Proposition. *If $x \in \mathfrak{m}$ is A -regular, then*

$$A \text{ is Gorenstein} \iff A/(x) \text{ is Gorenstein}.$$

Iterating shows that a complete intersection local ring—a quotient of a regular local ring by a regular sequence—is Gorenstein. In dimension zero, the criterion becomes

$$\dim_k \mathrm{Soc}(A) = 1, \quad \mathrm{Soc}(A) = (0 :_A \mathfrak{m}).$$

The socle formulation is especially useful after reducing a Cohen–Macaulay ring by a system of parameters.

80 Essential extensions and injective hulls

Definition. An extension $N \subseteq M$ is essential if every nonzero submodule of M meets N nontrivially. Every module N has an injective hull $E(N)$: an essential extension $N \subseteq E(N)$ with $E(N)$ injective. It is unique up to an isomorphism restricting to the identity on N .

If $N \subseteq M$ is essential and $M \subseteq E$ with E injective, then the extension of the identity on N embeds M into E . Zorn's lemma gives a maximal essential extension, which Baer's criterion shows is injective.

For a Noetherian ring, indecomposable injectives are precisely

$$E(A/\mathfrak{p}), \quad \mathfrak{p} \in \operatorname{Spec} A.$$

Every injective is a direct sum of these. Moreover

$$\operatorname{Ass} E(A/\mathfrak{p}) = \{\mathfrak{p}\},$$

and $E(A/\mathfrak{p})$ naturally carries an $A_{\mathfrak{p}}$ -module structure.

81 The injective hull of a residue field

Definition. Let $(A, \mathfrak{m}, k)$ be Noetherian local and $E = E_A(k)$. Then

$$\mathrm{Hom}_A(k, E) \cong k,$$

because the socle of E is the essential copy of k . Every element of E is annihilated by some power of $\mathfrak{m}$, so

$$E = \bigcup_{n \geq 1} (0 :_E \mathfrak{m}^n).$$

Each submodule $(0 :_E \mathfrak{m}^n)$ has finite length. Thus E is naturally a module over the completion $\widehat{A}$.

For $\mathfrak{p} \in \mathrm{Spec} A$, the module $E(A/\mathfrak{p})$ is obtained from the injective hull of $\kappa(\mathfrak{p})$ over $A_{\mathfrak{p}}$, viewed as an A -module. The indecomposable summands occurring in a minimal injective resolution of a module M are measured by the Bass numbers

$$\mu^i(\mathfrak{p}, M) = \dim_{\kappa(\mathfrak{p})} \mathrm{Ext}_{A_{\mathfrak{p}}}^i(\kappa(\mathfrak{p}), M_{\mathfrak{p}}).$$

If A is Gorenstein local of dimension d and $x_1, \dots, x_d$ is a parameter regular sequence, then $A/(x)$ is Artinian Gorenstein and its injective hull is itself. This is the zero-dimensional shadow of local duality.

82 Matlis duality for finite-length modules

Definition. Let $(A, \mathfrak{m}, k)$ be Noetherian local, $E = E_A(k)$, and

$$D(M) = \operatorname{Hom}_A(M, E).$$

Since E is injective, D is an exact contravariant functor. If M has finite length, then

$$\operatorname{length} D(M) = \operatorname{length} M, \quad M \xrightarrow{\sim} D(D(M)).$$

The proof starts with $M = k$, where $D(k) \cong k$, and proceeds by induction along a composition series. Exactness of D turns each short exact sequence around, and the five lemma proves biduality.

If M is Artinian, every finitely generated submodule has finite length, and

$$D(M) = \varprojlim_{N \subseteq M, \operatorname{length} N < \infty} D(N).$$

If M is finite, then $D(M)$ is Artinian. These observations lead to the complete-local form of Matlis duality.

83 Matlis duality over a complete local ring

Theorem (Matlis duality). *Let $(A, \mathfrak{m}, k)$ be a complete Noetherian local ring and $E = E_A(k)$. The functor $D = \operatorname{Hom}_A(-, E)$ gives mutually inverse anti-equivalences*

$$\{\text{finite } A\text{-modules}\} \longleftrightarrow \{\text{Artinian } A\text{-modules}\}.$$

For either class, the evaluation map $M \rightarrow D(D(M))$ is an isomorphism.

The endomorphism ring of the injective hull is

$$\operatorname{End}_A(E) \cong \widehat{A};$$

completeness therefore makes the double dual an A -module again without changing scalars. For arbitrary Noetherian local A , Matlis duality naturally lands over $\widehat{A}$.

Definition. An immediate consequence is that descending chains in an Artinian module dualize to ascending chains of finite modules. The theorem also turns finite direct sums into finite direct products and quotients into annihilator submodules.

84 Local cohomology

Definition. For an ideal $I \subseteq A$, define the I -torsion functor

$$\Gamma_I(M) = \{m \in M : I^n m = 0 \text{ for some } n\}.$$

It is left exact. Its right derived functors are the local cohomology modules

$$H_I^i(M) = R^i \Gamma_I(M).$$

Thus $H_I^0(M) = \Gamma_I(M)$, and every short exact sequence gives a long exact sequence in local cohomology. The functors depend only on $\sqrt{I}$ and commute with localization:

$$H_I^i(M)_{\mathfrak{p}} \cong H_{IA_{\mathfrak{p}}}^i(M_{\mathfrak{p}}).$$

They may also be computed as a direct limit,

$$H_I^i(M) \cong \varinjlim_n \operatorname{Ext}_A^i(A/I^n, M).$$

If M is I -torsion and J is injective, then maps $M \rightarrow J$ land in $\Gamma_I(J)$; this observation shows that Γ_I sends injectives to injectives and justifies the derived-functor construction.

85 The Čech complex

Proposition. If $I = (x_1, \dots, x_r)$, form the Čech complex

$$0 \rightarrow M \rightarrow \bigoplus_i M_{x_i} \rightarrow \bigoplus_{i < j} M_{x_i x_j} \rightarrow \cdots \rightarrow M_{x_1 \cdots x_r} \rightarrow 0.$$

With the alternating localization maps, its cohomology computes local cohomology:

$$H_I^i(M) \cong H^i(\check{C}^\bullet(x; M)).$$

One obtains the complex as a direct limit of Koszul cochain complexes on $x_1^n, \dots, x_r^n$. Hence

$$H_I^i(M) = 0 \quad (i > r).$$

For a principal ideal,

$$0 \rightarrow H_{(x)}^0(M) \rightarrow M \rightarrow M_x \rightarrow H_{(x)}^1(M) \rightarrow 0$$

and higher groups vanish.

If $(A, \mathfrak{m})$ is local and M is finite, the least i for which $H_{\mathfrak{m}}^i(M) \neq 0$ is $\text{depth } M$. This agrees with the earlier Ext characterization of depth.

86 Grothendieck vanishing and nonvanishing

Let $(A, \mathfrak{m})$ be Noetherian local and M finite of dimension d . Then

$$H_{\mathfrak{m}}^i(M) = 0 \quad (i > d), \quad H_{\mathfrak{m}}^d(M) \neq 0 \quad (M \neq 0).$$

The vanishing theorem follows by induction on dimension. Remove $\Gamma_{\mathfrak{m}}(M)$, choose an M -regular element $x \in \mathfrak{m}$, and use the long exact sequence attached to $0 \rightarrow M \xrightarrow{x} M \rightarrow M/xM \rightarrow 0$. Localization and induction control the remaining support.

For nonvanishing, reduce to a cyclic quotient $A/\mathfrak{p}$ of maximal dimension, use Noether normalization in the complete case, and compare with the top Čech class of a system of parameters. The result shows that local cohomology detects dimension at its highest nonzero degree:

$$\dim M = \sup\{i : H_{\mathfrak{m}}^i(M) \neq 0\}.$$

87 Local duality

Let $(A, \mathfrak{m}, k)$ be a complete regular local ring of dimension d , let $E = E_A(k)$, and write $D(-) = \operatorname{Hom}_A(-, E)$. Then for every finite A -module M ,

Theorem (Local duality).

$$D(H_{\mathfrak{m}}^i(M)) \cong \operatorname{Ext}_A^{d-i}(M, A).$$

Equivalently,

$$H_{\mathfrak{m}}^i(M) \cong D(\operatorname{Ext}_A^{d-i}(M, A)).$$

For $M = A$, the theorem says $H_{\mathfrak{m}}^i(A) = 0$ for $i \neq d$ and

$$H_{\mathfrak{m}}^d(A) \cong E.$$

Choose a regular system of parameters $x_1, \dots, x_d$. The top Čech cohomology is the direct limit of $A/(x_1^n, \dots, x_d^n)$ with transition maps multiplication by $x_1 \cdots x_d$. Matlis duality identifies its dual with the inverse limit, namely A by completeness. This proves the assertion for A .

Both sides of local duality are cohomological delta functors in M . Resolve a finite module by finite free modules and use the result for free modules; dimension shifting gives the displayed isomorphisms in every degree.

Corollary. *For finite M ,*

$$\operatorname{depth} M = \inf\{i : H_{\mathfrak{m}}^i(M) \neq 0\}, \quad \dim M = \sup\{i : H_{\mathfrak{m}}^i(M) \neq 0\}.$$